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Long-Term Water Balance Dynamics in a Brazilian Neotropical Forest: Insights from Seven Years of Continuous Observation

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January 28, 2025

Dear Editor:

I am writing to submit the manuscript “**Long-Term Water Balance Dynamics in a Brazilian Neotropical Forest: Insights from Seven Years of Continuous Observation**” to the Editorial Board of *Agricultural and Forest Meteorology* for publication. This study is the first carried out in Brazil in which seven complete hydrological years of monitoring were used to understand water fluxes in an Atlantic Forest remnant in southeast Brazil. We demonstrated that without these longer study periods, the results have been little conclusive as different behaviors of the climate are not considered. In our study, the forest system was subjected to severe droughts, abnormal droughts and wet/average hydrological years. Our results have great potential for application in studies on the ecosystem services this very sensitive biome in Brazil offers.

I will be the corresponding author and can be reached by e-mail at crmello@ufla.br.

Thank you very much for your disposition.

Best wishes



Prof. Carlos R. de Mello - Corresponding author

Highlights

- Seven hydrological years were used to assess the water balance in a seasonal tropical forest
- In severe drought years, evapotranspiration corresponded to more than 94% of precipitation
- Deep percolation was 4% of precipitation in drought years and 25.4% in abnormal/wet years
- Atmosphere energy predominates on evapotranspiration controls in wet years and soil-water availability in drought conditions

Long-Term Water Balance Dynamics in a Brazilian Neotropical Forest: Insights from Seven Years of Continuous Observation

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Abstract

The Atlantic Forest ecosystems offer numerous ecosystem services. However, UNESCO regards it as one of the most fragile biosphere reserves globally due to anthropogenic pressures in Brazil that threaten its integrity. This study evaluated the water balance relationship in an Atlantic Forest fragment using hydrometeorological data collected over seven hydrological years (2014-2020) of continuous observation of the hydrometeorological elements and soil moisture. The focus was on analyzing evapotranspiration and its biophysical controls, the canopy rainfall interception, soil moisture, and percolation over time. On average, canopy rainfall interception accounted for 21% of gross precipitation (GP), and evapotranspiration constituted an average of 79.8% (± 11.6) of GP, peaking at 97.6% during the driest hydrological year of the study period. Evapotranspiration processes were primarily governed by stomatal and soil water storage, mainly during dry periods. We observed that the severe droughts have severely impacted the water balance in the forest, with an ET/GP ratio greater than 95% in 2013/2014 and 2016/2017, the driest hydrological years observed in the region in the last 100 years. Deep percolation showed significant variation, ranging downward and upward during rainy and dry periods, respectively, driven by increased evapotranspiration and a marked reduction in GP and soil moisture. The water balance components highlighted that the Atlantic Forest is a crucial biome and a lifeline for providing ecological functions, particularly in water control and storage, even during dry periods.

Keywords: soil water storage, biophysical controls, rainfall interception, evapotranspiration, Atlantic Forest.

49 **Introduction**

50 The Atlantic Forest Biome, home to more than 60% of Brazil's urban population, is under
 51 considerable pressure and fragmentation. This degradation has led to vegetation losses and adverse
 52 impacts on climate and hydrological regimes, underlining the urgent need for conservation. Taffarello
 53 et al. (2017) and Macedo et al. (2021) have emphasized the crucial role of Atlantic Forests in providing
 54 water services, mitigating persistent droughts, and reducing natural hazards in Brazil. UNESCO has
 55 recognized the Atlantic Forest as one of the critical biosphere reserves worldwide, given its diversity,
 56 endemism, and environmental significance (Myers et al., 2000). Currently, the biome retains only 28%
 57 of its original coverage, with 91% of this area experiencing continuous fragmentation. In addition, 46%
 58 of the remaining forest is subject to a strong edge effect (Macedo et al., 2021; Poorter et al., 2016),
 59 exhibiting low resilience (Tambosi et al., 2014). These facts highlight the pressing need to preserve the
 60 remaining biodiversity and ecohydrological functions of the Atlantic Forest Biome (Silva et al., 2020;
 61 Rezende et al., 2018). Studying the hydrological dynamics and processes of this ecosystem is crucial
 62 and urgent to support strategies for the conservation and sustainable management of the Atlantic Forest.
 63 Among the hydrological processes on forest water balance, evapotranspiration is vital in linking the
 64 critical zone and atmosphere systems, playing a crucial role in the water and energy cycles (Guauque-
 65 Mellado et al., 2022; Cabral et al., 2015; Costa et al., 2010; Marques et al., 2020; Tan et al., 2019).
 66 Therefore, advancing our understanding of processes and impacts on the water balance of the Atlantic
 67 Forest is essential to enhance our understanding of their role in the regional climate dynamics, water
 68 quantity and quality, and improve management of these ecosystems.

69 The hydrological processes in the Atlantic Forest are affected by a marked climatic seasonality
 70 characterized by dry winters and wet summers, resulting in its semi-deciduous nature (Freitas et al.,
 71 2020). Species in semi-deciduous ecosystems typically self-regulate the physiological process of
 72 transpiration, shedding foliage during rainless periods to avoid water stress (Marques et al., 2020). This
 73 aspect should be considered in studies of forest water balance, as it can affect forest water storage
 74 capacity throughout the year (Mello et al., 2019). Several studies have examined the evapotranspiration
 75 of Atlantic Forest biomes in short-term studies spanning less than two years (Mello et al., 2019),

offering only partial insights into this ecosystem's dynamics of evapotranspiration and water pathways. In addition, the semi-deciduous nature of the Atlantic Forest, marked by climatic seasonality, presents a unique challenge for understanding its water balance. The interaction between meteorological variables, vegetation composition, phenology, and soil water attributes on water balance in Atlantic Forest areas remains poorly understood. Therefore, further knowledge of evapotranspiration partition in this ecosystem is crucial, especially under the persistence of heat waves that are more prominent in climate change.

This study aims to evaluate the water balance components of an Atlantic Forest remnant based on hydrometeorological data spanning seven hydrological years (October 2013 to September 2020). Specifically, we seek to (i) identify the dynamics and partitioning of the rainfall interception and its influences on soil water storage; (ii) estimate evapotranspiration using meteorological data and identify its main drivers; and (iii) estimate the elements of the water balance using a unique hydrological dataset that encompasses both dry and wet years.

Materials and Methods

Site description

The studied area (Figure 1) is in Southeast Brazil, in the coordinates 21°13'40''S and 44°57'50''W, with an average altitude of 925 meters. It consists of a seasonal semideciduous mountain forest (SSMF) remnant of 6.35 ha. The soil is a Red Latosol distributed over a slightly undulated relief with slopes < 15% (Junqueira et al., 2017). The Köppen climate of the region is mesothermal subtropical, i.e., Cwa, with well-defined precipitation seasonality (Junqueira Junior et al., 2019): wet (October to March) and dry (April to September) periods. The mean annual precipitation is 1461.8 mm, with 84% concentrated in the wet season. The mean annual temperature is 20.3°C, varying from 16.9°C in July to 22.8°C in February (INMET, 2022).

The experimental area has been periodically monitored regarding vegetation parameters (forest inventory) since 1994, measuring species growth and identifying tree mortality. At first, it was

inventoried by 6527 individuals with a diameter at breast high (DBH) > 5 cm by Oliveira-Filho et al. (1994). However, 5609 individuals were registered in the last forest inventory (2017), distributed among forty-eight botanical families and 184 species. The most representative families are *Annonaceae* (830), *Fabaceae* (769), *Lauraceae* (720), *Rubiaceae* (671), *Myrtaceae* (510), *Meliaceae* (431), and *Melastomataceae* (363). The most frequent species are: *Xylopia brasiliensis* (670), *Copaifera langsdorffii* (517), *Amaioua intermedia* (497), *Trichilia emarginata* (383), *Miconia willdenowii* (282), *Myrcia splendens* (262), and *Casearia arborea* (195).

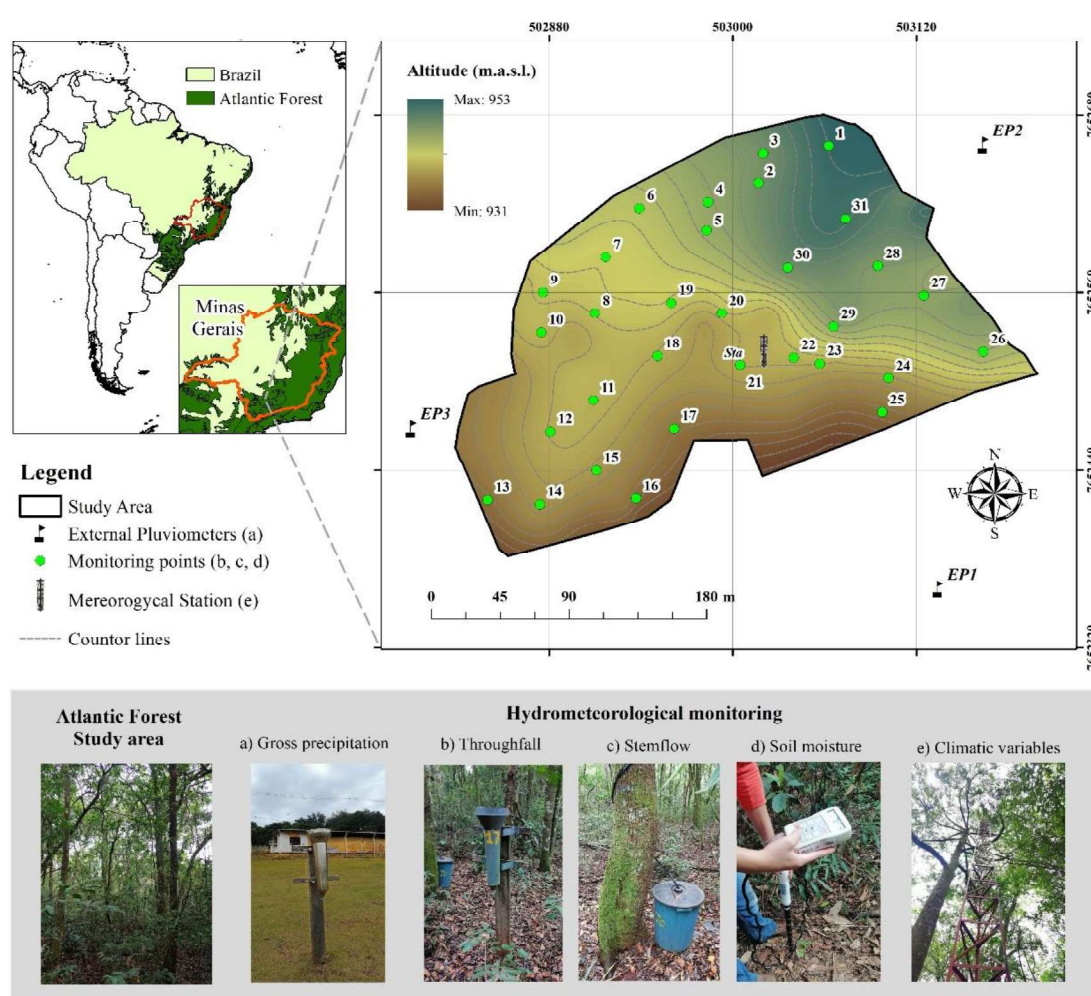


Figure 1. Geographical location and instrumentation used for hydrometeorological monitoring in the SSMF, southeastern Brazil.

The forest is in a stage of transition (Rodrigues et al., 2021), with 87.68% of individuals in minor classes of DBH, i.e., 5-15 cm (68.41%) and 15-25 cm (19.29%). The other classes correspond to

12.32%, in which the 25-35 cm class comprehends 8.34% and > 35 cm, 3.97%. The forest canopy's average height is 10.2 m, with a few dispersed individuals higher than 20 m (Junqueira et al., 2019). The canopy stratification has four levels: (i) a layer of emerging trees (>20 m) dispersed in the area; (ii) a layer of the main canopy (10-15 m); (iii) a layer of shrub and young trees (<10 m); and (iv) an herbal cover over the soil. Openings in the canopy are caused by fallen and dead trees that have completed their cycle, which modifies the species' dynamic and the remnant's structure.

Monitoring framework

Gross precipitation (GP), i.e., the incoming rainfall, was measured between October 2013 and September 2020. In this period, 644 rainy events were monitored. At first, between 2013 and 2017, the measurement was carried out through two procedures: (i) automatic monitoring, performed every hour using an 8-inch tipping-bucket rain gauge that covers 324.3 cm² (TE525-L, Texas Electronics Inc., Dallas, TX, EUA); and (ii) daily manual monitoring, performed using a totalizing-type “Ville de Paris” rain gauge with a collected area of 378.5 cm². Both rain gauges were installed at a 22 m high Micrometeorological Observation Tower (MOT) near the study area's center (Figure 1). From August 2017, three other totalizing-type rain gauges were installed surrounding the SSMF to complement the monitoring system and improve the spatial representativeness of GP (Figure 1a).

In the MOT (Figure 1e), micrometeorological sensors were installed at two heights, one at the top of the forest canopy (22 m high) and another at 2 m above ground. Each level contains sensors for relative humidity (RH) and temperature (T_a) measurements (Model S-THB-M002). A pyranometer set (measuring incoming and outgoing radiation) (Model S-LIB-M003) for net radiation (R_n), a barometric pressure sensor (Model CS100), and an anemometer (Model 03002) for wind speed and direction were also installed. The sensor data were stored hourly in a CR10x programmable data logger (Campbell Scientific, Logan, UT, USA).

Thirty-one points were randomly selected inside the SSMF to measure throughfall (TF), stemflow (Sf), and soil moisture (θ) up to 1.0 m depth (Figure 1). These points are approximately spaced

40 m from each other. TF measurement (Figure 1b) was carried out with a “Ville de Paris” rain gauge installed 1.5 m above the forest floor to avoid splash-in (Junqueira et al., 2019). Data collection was performed four hours after the rainfall had ceased (and on the following day when it ceased late afternoon) to guarantee that the canopy had wholly drained and prevented bias.

Sf was measured in one representative tree (Figure 1c) for each monitoring point. The individuals were selected according to localization, abundance, and dominant species (Terra et al., 2018). The collectors were constructed with polyethylene hose set in the tree trunks in a spiral scheme, with silicone applied between the collector and tree to avoid leaking (Mello et al., 2023). The stemflow was drained into a container and collected simultaneously to TF.

In each monitoring point, permanent access tubes were installed (Figure 1d) to measure the soil moisture (m^3m^{-3}) in five depths (0.10 m, 0.20 m, 0.30 m, 0.40 m, and 1.0 m) at ~ 15 -day intervals. The measurement was carried out with a multi-sensor capacitance probe (model PR2/6, Delta-T Devices Ltd.).

The leaf area index (LAI in m^2m^{-2}) was measured monthly at the 31 monitoring points using a previously calibrated and validated Sun Scan zeptometer system (Model SS1-STD3, Delta-T Devices Ltd., Cambridge, UK). These LAI datasets were used to parametrize the heat flux from the soil considered in the Penman-Monteith model for evapotranspiration estimates.

Water balance calculation

The water balance of the forest remnants was monthly computed according to Jones et al. (2017):

$$\frac{\Delta S}{\Delta t} = \text{GP} - \text{ET} - \text{DP} - \text{R} \quad (1)$$

Where GP (mm) is the gross precipitation, ET (mm) is the evapotranspiration, S (mm) is the soil water storage integrated over the root zone (1.0 m in this study), DP (mm) is the deep-water percolation

leaving the root zone, R is the surface runoff (mm), and Δt is the interval adopted for the water balance computation (monthly). GP was calculated by averaging the precipitation of the external rain gauges. $\frac{\Delta S}{\Delta t}$ was calculated for each one of the 31 sets of monitoring as a function of the soil moisture readings interval. Surface runoff was negligible as the topography of the study area is smooth undulated to flat, making this component too small (in a recent runoff study in this area, we observed approximately 1.5 mm in the 2022/2023 hydrological year).

Although tropical forests can uptake water from deeper layers during drought conditions (Markewitz et al., 2010), this amount is of minor contribution to total evapotranspiration as it is restricted to short periods (Broedel et al., 2017). Thus, the root water uptake was considered up to 1.0 m depth since we observed a decrease in thinner root abundance during the installation of the monitoring system, concentrated between 0.50 and 1.0 m. S was calculated as follows:

$$S = \sum_{i=1}^n \left(\frac{\theta_i + \theta_{i+1}}{2} \cdot h \right) \quad (2)$$

$$\frac{\Delta S}{\Delta t} = \frac{S_{(t+1)} - S_{(t)}}{\Delta T} \quad (3)$$

where θ_i e θ_{i+1} are, respectively, soil moisture at depth i and $i + 1$ ($m^3 m^{-3}$), n is the number of layers (0–0.10 m; 0.1–0.2 m; 0.20–0.30 m; 0.30–0.40 m; 0.40–1.0 m), h is the thickness of each layer (mm), and t is the consecutive date of each measurement.

The evapotranspiration can be calculated using the water balance equation (ET_{WB}) in periods of low rainfall, i.e., when percolation can be neglected (Tomasella et al., 2008; Mello et al., 2019):

$$\frac{ET_{(WB)}}{\Delta T} = \frac{(TF + Sf) - \Delta S_{(0-1.0\text{ m})}}{\Delta T} \quad (4)$$

The canopy interception (C) was calculated considering the average canopy rainfall interception (mm) using the 31 observed points, obtained through the equation described by Ghimire et al. (2014) and Junqueira et al. (2019):

$$\frac{C}{\Delta T} = \frac{GP - (TF + Sf)}{\Delta T} \quad (5)$$

Percolation was estimated as a residue of the water balance from Equation 1 in the rainy period. According to Jones et al. (2017), this simplification implies that when GP exceeds ET, DP tends to be positive, meaning that the water is leaving the root zone, i.e., percolation. On the other hand, when ET exceeds GP, DP is negative, and water enters the root zone (capillary rise).

Since forest evapotranspiration relies on soil water availability, we calculated the relative extractable water (REW) (Allen et al., 1998):

$$REW = \frac{(\theta - \theta_{wp})}{(\theta_{fc} - \theta_{wp})} \quad (6)$$

where θ , θ_{fc} , and θ_{wp} , are the water volumetric content, field capacity, and wilting point obtained by Rodrigues et al. (2020).

Parameterization of the evapotranspiration model

During the rainy season (October – March), the dependent variable of the water balance was the deep percolation (DP), i.e.:

$$\frac{DP}{\Delta t} = \frac{TF + Sf - ET_M - \Delta S}{\Delta t} \quad (7)$$

We modeled ET for SSMF using equation 8. For that, it was also performed a parameterization of the ET encompassing soil evaporation and plant transpiration into a single evaporative element process (Ershadi et al., 2015), which is used to describe the physiological and environmental conditions (Monteith, 1965; Kelliher et al., 1995). ET is estimated by:

$$ET = \frac{\Delta \cdot (R_n - G) + \rho_a \cdot c_p \cdot g_a \cdot (VPD)}{\lambda \cdot \left[(\Delta + \gamma) \cdot \left(1 + \frac{g_a}{g_s} \right) \right]} \quad (8)$$

which ET is the modeled evapotranspiration (mm d⁻¹); λ is the latent heat of water vaporization (MJ kg⁻¹); Δ is the declivity of saturation vapor pressure curve (kPa °C⁻¹); R_n is the net radiation (MJ m⁻² d⁻¹); G is the heat flux in the soil (MJ m⁻² d⁻¹); γ is the psychrometric coefficient (kPa °C⁻¹); ρ_a is the air density (kg m⁻³); VPD is the water vapor pressure deficit (kPa); C_p is the specific heat at constant pressure (J kg⁻¹ °C⁻¹) (approximately 1,000 J kg⁻¹ °C⁻¹); g_a is the aerodynamic conductance (m s⁻¹), and

g_s is the canopy conductance (m s^{-1}). G was obtained through the empirical model of Choudhury et al. (1987), which relates LAI with R_n as:

$$G = 0.079 \cdot [(-0.24 \cdot \text{LAI})] \cdot R_n \quad (9)$$

The aerodynamic conductance (g_a) was determined by focusing on Monin-Obukhov (Garrat and Hicks, 1973; Loescher et al., 2005) theory, which is based on vertical wind profile. Such modeling was performed daily, and the measurement was restricted to neutral conditions of balance in the atmosphere (Allen et al., 1998):

$$g_a = \frac{k^2 \cdot u(Z)}{\left\{ \ln \left[\frac{(Z-d)}{z_{0M}} \right] \right\}^2} \quad (10)$$

“ u ” is the wind speed (m s^{-1}) in the Z high (m) (22 m in MOT), d is the zero-plane displacement (m), z_{0M} is surface roughness (m), and k is the von Kármán constant (0.41). Values of z_{0M} and d considered were $(0.1) \cdot h_c$ e $(2/3) \cdot h_c$, respectively; h_c corresponds to the canopy average high, approximately 10.2 m.

There is a relevant difficulty in the canopy conductance parameterization, and several models with empirical approaches that link various parameters that influence the exchange of gases in the stomata have been proposed (Tan et al., 2019). In this study, g_s was obtained by the Penman-Monteith (Eq. 8), using indirect values of λE extracted from the energy balance with residual focus (Fischer et al., 2013; Su, 2002):

$$R_n - G - H = \lambda E \quad (11)$$

This involves the measurement of sensible heat flux H ($\text{MJ m}^{-2} \text{d}^{-1}$) using the "Inversing Bulk Transfer Equation" method (based on the Monin-Obukhov similarity theory) (Norman et al., 1995). This method has been applied in different research with micrometeorological approach and remote sensors for the energy flux evaluation (Kalma et al., 2008; Silberstein et al., 2003), being H expressed as:

$$H = c_p \cdot \rho_a \cdot g_a \cdot (T_1 - T_2) \quad (12)$$

T_1 is the air temperature near an aerodynamic surface at canopy height ($^{\circ}\text{C}$); T_2 is the temperature near the surface ($^{\circ}\text{C}$). T_1 is the daily average temperature measured on MOT, while T_2 is the daily average temperature measured inside the forest at 2.0 m above the ground.

To support the knowledge of the physical controls of evapotranspiration in SSMF, the decoupling coefficient (Ω) and Priestley-Taylor coefficient (α) were calculated. Thus, the evapotranspiration sensitivity to changes in canopy characteristics and local atmosphere conditions can be evaluated through the Ω (McNaughton and Jarvis, 1983):

$$\Omega = \frac{1}{1 + \left[\frac{\gamma}{\Delta + \gamma} \right] \cdot \left[\frac{g_a}{g_s} \right]} \quad (13)$$

According to Ma et al. (2015) and Marques et al. (2020), Ω varies from 0 to 1. Values close to 0 indicate that evapotranspiration is mainly controlled by stomata closure. In contrast, values near to 1 suggest that energy balance is the limiting factor of evapotranspiration.

Furthermore, to assess if atmospheric demand or soil moisture supply controls evapotranspiration, the α parameter was calculated (Priestley and Taylor, 1972). This is equivalent to the ratio between evapotranspiration and the equilibrium evapotranspiration (E_{eq}), i.e., depends on net radiation and temperature:

$$\alpha = \frac{\lambda E}{\lambda E_{eq}} = \frac{\lambda E}{\left[\frac{(R_n - G) \Delta}{(\Delta + \gamma)} \right]} \quad (14)$$

where λE is the modeled latent heat flux (Eq. 11). R_n , G , Δ and γ were described previously. Values of $\alpha < 1$ denote ecosystem limitations to the water supply; $\alpha > 1$ represents a humid ecosystem, i.e., there are no water limitations, and only the available energy limits evapotranspiration (Marques et al., 2020).

Spatial variability of $ET_{(WB)}$ and precision statistics

The $ET_{(WB)}$ values were mapped in the SSMF, referencing each hydrological year's dry season (from April to September) between 2014 and 2020. The spatial variability was evaluated through

geostatistical procedures (ordinary kriging method) and the coefficient of variation (CV) among sampled points of the water balance (Figure 1). The interpolations were performed in ArcGIS (version 10.8) using the exponential semivariogram model, which is widely used for hydrological and soil physical attributes mapping (Junqueira et al., 2017; Kamali et al., 2015; Rodrigues et al., 2020).

A comparative analysis was performed between modeled evapotranspiration (ET) and evapotranspiration (ET_{WB}) obtained from water balance during months that did not present significant precipitation. The slope hypothesis (line 1:1) and its respective coefficient of determination (R^2) were evaluated. The performance of ET to simulate evapotranspiration in the SSMF was evaluated by Root Mean Square Error (RMSE) and Nash-Sutcliffe (NS) coefficient:

$$RMSE = \sqrt{\frac{1}{N} \left\{ \sum_{i=1}^N (P_i - O_i) \right\}^2} \quad (13)$$

$$NS = 1 - \frac{\sum_{i=1}^N (P_i - O_i)^2}{\sum_{i=1}^N (O_i - O_m)^2} \quad (14)$$

where P_i and O_i are simulated and observed evapotranspiration at time i , respectively; N is the number of simulated and observed pair values; and O_m is the observed average values. Values of RMSE close to zero and NS values close to 1 indicate a good model fit.

Results

Meteorological conditions observed in the studied period

The daily average GP was 3.11 mm d⁻¹ during the studied years, showing substantial seasonal variations (Figure 2a). Notably, two GP events exceeding 100 mm d⁻¹ were recorded in 2019 (November and December) and one in 2020 (February). The daily average values for the hydrological years are in Table S1 (Supplementary). The average air temperature (T) above the forest canopy (Figure 2b) was 21.9°C in the wet season and 18.5°C in the dry season, approximately 0.05°C lower and 2.51°C higher than the long-term average, respectively (INMET, 2022). Inside SSMF, T ranged from 21.9°C in the wet season to 15.8°C in the dry season. Solar radiation (Rs) (Figure 2c) presents daily seasonal variation

with higher radiation intensity in the wet season (18.47 MJ m^{-2}) and the lowest in the dry season (15.15 MJ m^{-2}). The average vapor pressure deficit (VPD) was 0.58 kPa and 0.54 kPa in the dry and wet seasons, respectively (Figure 2d).

Figure 2e presents the relative extractable water (REW) at five soil depths (0.1 m , 0.2 m , 0.3 m , 0.4 m , and 1.0 m). During the wet season, frequent rainfall maintains REW close to 1 across all depths, fulfilling the field capacity. In contrast, during the dry season, the forest's water uptake causes REW to decrease gradually, especially in the 0.1 , 0.2 , and 0.3 m depths, where average values decrease to 0.24 , 0.44 , and 0.48 , respectively. However, at the 1.0 m depth, REW remains approximately 0.7 to 0.8 .

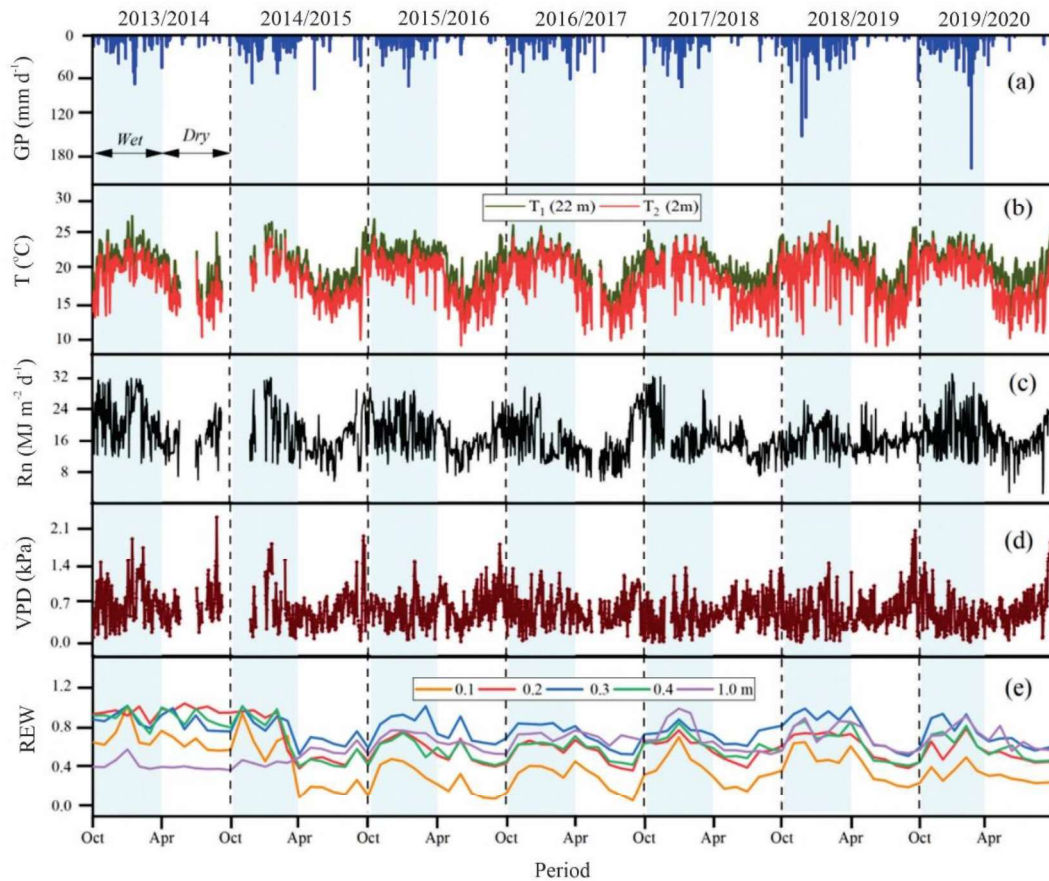
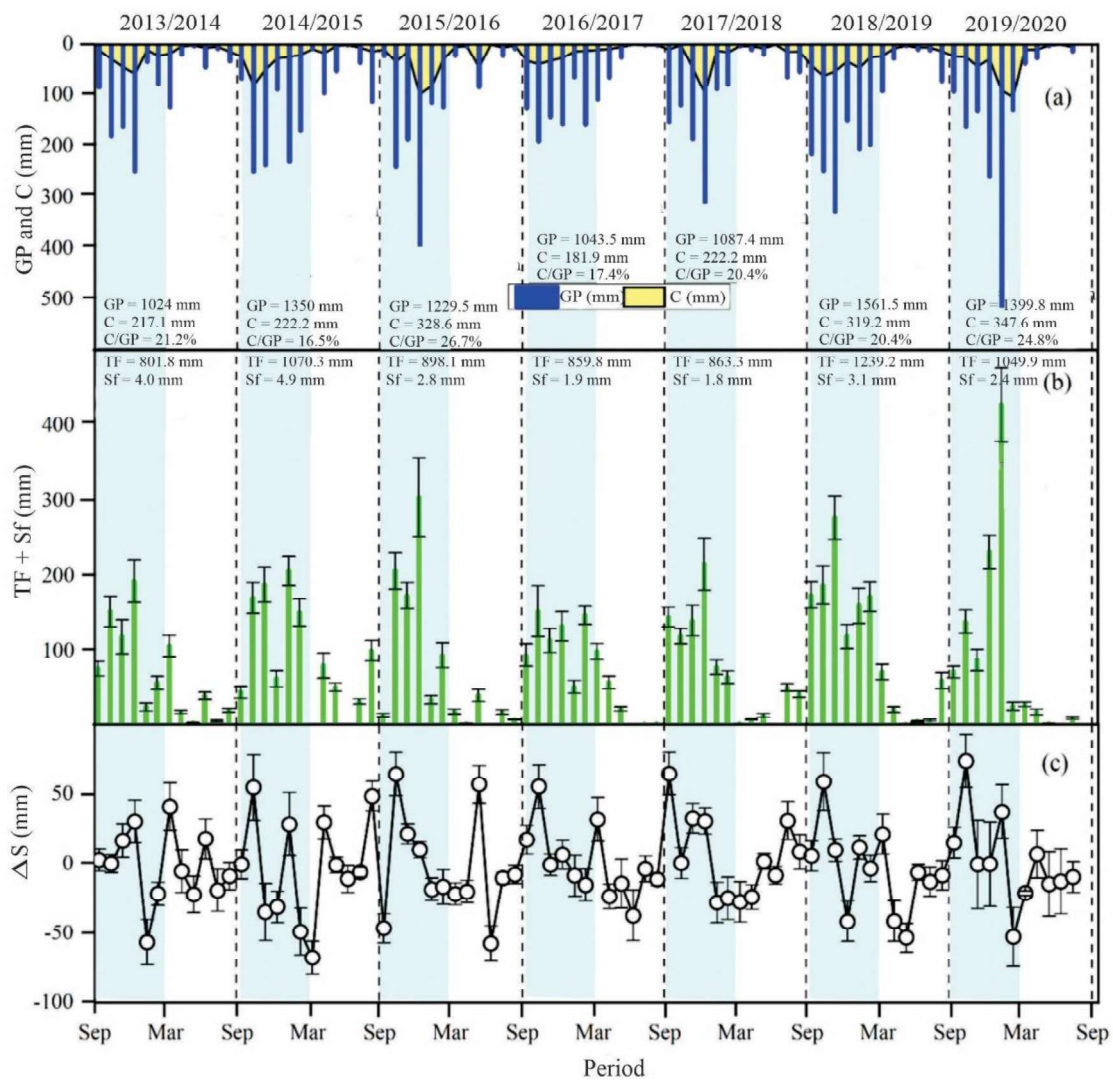


Figure 2. Average daily (a) GP, (b) air temperature (T), (c) global solar radiation (R_s), (d) vapor pressure deficit (VPD), and (e) relative extractable water (REW) in the $0\text{--}1.0\text{m}$ soil layer. Shaded columns represent wet seasons.

295 *Precipitation partitioning and soil water storage in the studied period*

296 The GP values for each hydrological year were as follows: 1024 mm (2013-2014), 1350.1 mm
 297 (2014-2015), 1229.5 mm (2015-2016), 1043.5 mm (2016-2017), 1087.4 mm (2017-2018), 1561.5 mm
 298 (2018-2019), and 1399.8 mm (2019-2020). In these seven years of observation, the average GP was
 299 1242 mm (± 204.0 mm). Figure 3 presents the studied forest's monthly water balance throughout the
 300 hydrological years (September – October).



301

302 Figure 3. Annual and monthly gross precipitation (GP) and canopy rainfall interception (C) (a), annual
 303 and monthly effective (or net) precipitation (NP = TF + Sf), with respective standard deviations (vertical
 304 bars) (b), monthly soil water storage variation in the 0-1.0 m control layer (ΔS), with respective standard
 305 deviations (vertical bars) (c). Shaded columns represent wet seasons.

C had a proportional variation with rainfall amount across the studied years, averaging 21.06% ($\pm 3.67\%$). In the hydrological year of 2019-2020, C increased to 24.8% due to a more significant rainfall amount. The net precipitation (TF + Sf) represented 78.24% ($\pm 7.2\%$) of GP, ranging from 805.8 mm (2014-2015) to 1242.3 mm (2018-2019). TF in the wet season ranged from 611.4 mm to 1080.3 mm (an average of 818.7 mm; 78.2% of GP). In the dry season, TF varied from 77.1 mm to 257.7 mm, averaging 149.6 mm (76.7% of GP). The annual average Sf was 2.58 mm (from 1.5 mm to 3.9 mm) during wet and 0.4 mm (from 0.1 mm to 1.0 mm) during dry seasons; on average, Sf was 0.24% (± 0.10 mm) of GP in the studied period.

The monthly soil water storage variation (ΔS) stood out with a marked predominance of positive values (i.e., increased soil water storage) in the wet season (Figure 3c), with an average of 28 mm. However, at the end of the 2013-2014 and 2014-2015 hydrological years, ΔS presented -28.5 mm and -32.2 mm, respectively. In the dry seasons, the lack of rainfall presented a higher recession of soil water storage by water consumption in the ecosystem, averaging -41.4 mm.

Figure 4 presents the mathematical relationships between TF, Sf, C/GP and ET hydrological variables, and GP. TF and Sf showed a linear relationship (Figure 4a, b), following the seasonal pattern of GP. C/GP and GP had a log relationship (Figure 4c), and ET presented an exponential relation (Figure 4d).

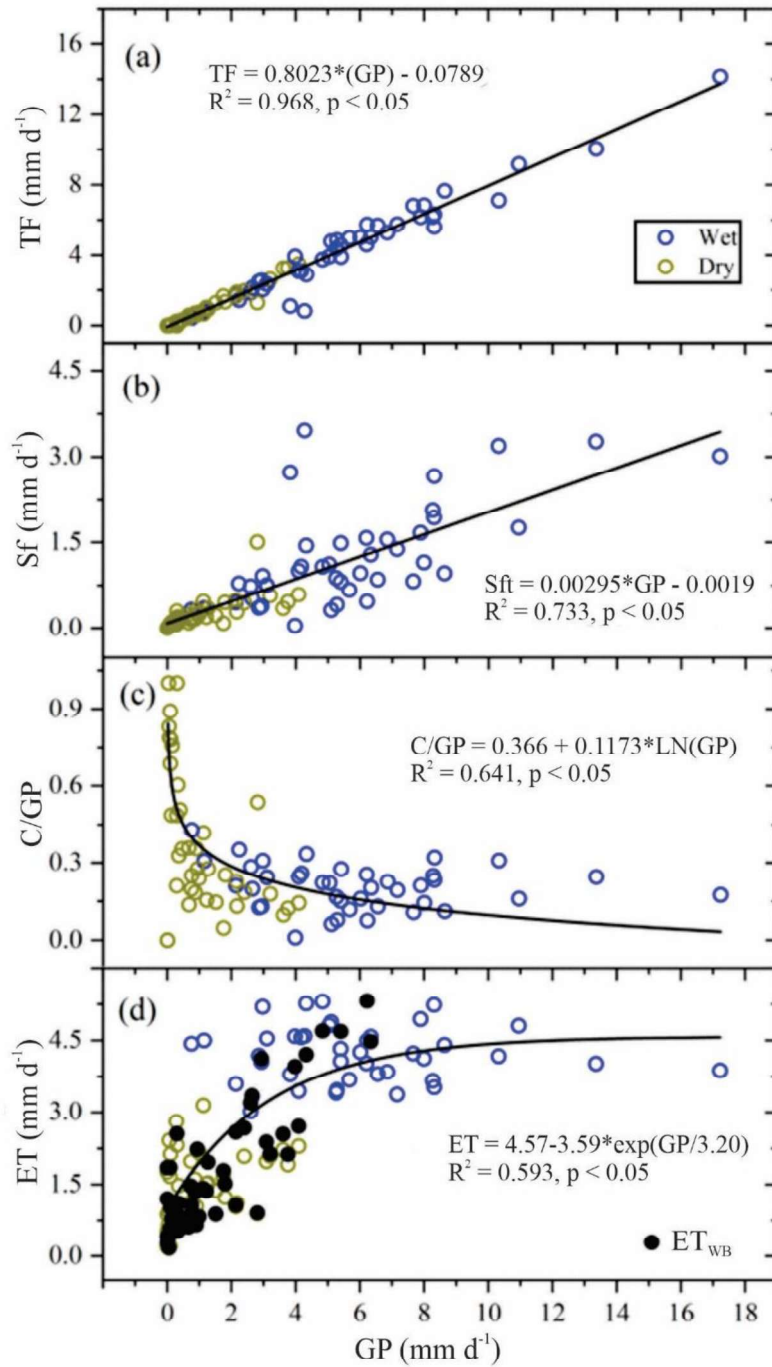
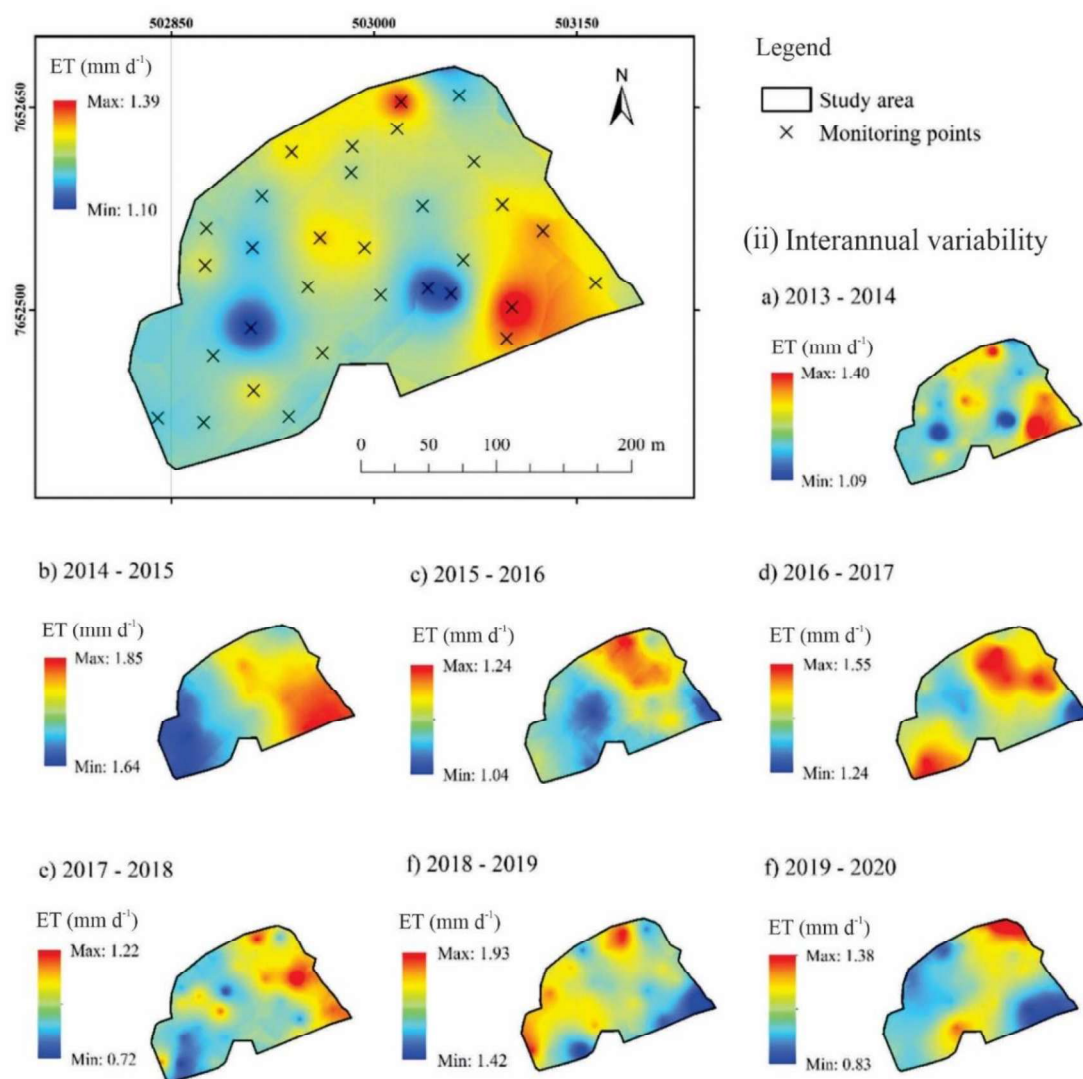


Figure 4. Relationships between daily average gross rainfall (GP): (a) throughfall (TF); (b) stemflow (Sf); (c) C/GP ratio (Sf); and (d) actual evapotranspiration (ET_{WB}).

329 *Evapotranspiration modeling*

330 Figure 5 presents the daily average evapotranspiration from the monthly water balance
 331 performed during the dry season (i.e., April-September, from 2014 to 2020). Evapotranspiration varied
 332 between 0.90 and 1.75 mm d⁻¹, with the hydrological years 2017-2018 and 2014-2015 recording the
 333 minimum and maximum values, respectively. The coefficient of variation (CV) varied from 5% to
 334 14.17%, with the lowest values occurring in the driest years.

(i) Multiannual variability of actual evapotranspiration (ET)



335

336 Figure 5. Dry period daily average evapotranspiration: (i) multiannual variability (average from 2014
 337 to 2020); (ii) interannual variability.

The temporal behavior of the modeled daily evapotranspiration, energy fluxes, Priestley-Taylor coefficient (α), canopy conductance (g_s), and the decoupling factor (Ω) are displayed in Figure 6. The R_n values showed clear seasonality, with the highest values in the wet season ($16.21 \text{ MJ m}^{-2} \text{ d}^{-1}$ on average) and the lowest in the dry season ($9.21 \text{ MJ m}^{-2} \text{ d}^{-1}$ on average) (Figure 6a). The heat flux in the soil is near zero, corresponding on average to $0.43 \text{ MJ m}^{-2} \text{ d}^{-1}$ in wet seasons and $0.18 \text{ MJ m}^{-2} \text{ d}^{-1}$ in dry seasons. H also presents a seasonality fluctuation, with a daily average value of $6.16 \text{ MJ m}^{-2} \text{ d}^{-1}$ in the dry season, whereas in the wet season, $5.37 \text{ MJ m}^{-2} \text{ d}^{-1}$. On the other hand, higher values of λE (Figure 6b) occurred in the wet season, with an average of $10.28 \text{ MJ m}^{-2} \text{ d}^{-1}$ (equivalent to 4.20 mm d^{-1}) and $3.11 \text{ MJ m}^{-2} \text{ d}^{-1}$ (equivalent to 1.27 mm d^{-1}) for dry season.

Regarding R_n , λE corresponded to 60% of it in the study period. The Priestley-Taylor coefficient (α) presented values > 1 in the wet season, indicating that evapotranspiration is mainly controlled by atmospheric demand (Figure 6c) in this period. An average value of 0.51 in the dry seasons means that soil moisture also controls evapotranspiration, but atmospheric demand still plays a significant role in this period. In addition, g_s (Figure 6d) presented the same seasonal pattern as evapotranspiration, with an average in the wet season of 3.47 mm s^{-1} and, in the dry season, 0.85 mm s^{-1} . The seasonal pattern of Ω (Figure 6e) was similar to g_s . The low values indicate a substantial control of evapotranspiration through g_s in dry (0.10) and wet (0.35) seasons. Table S2 (supplementary) details the daily average values for these components throughout the seven hydrological years studied.

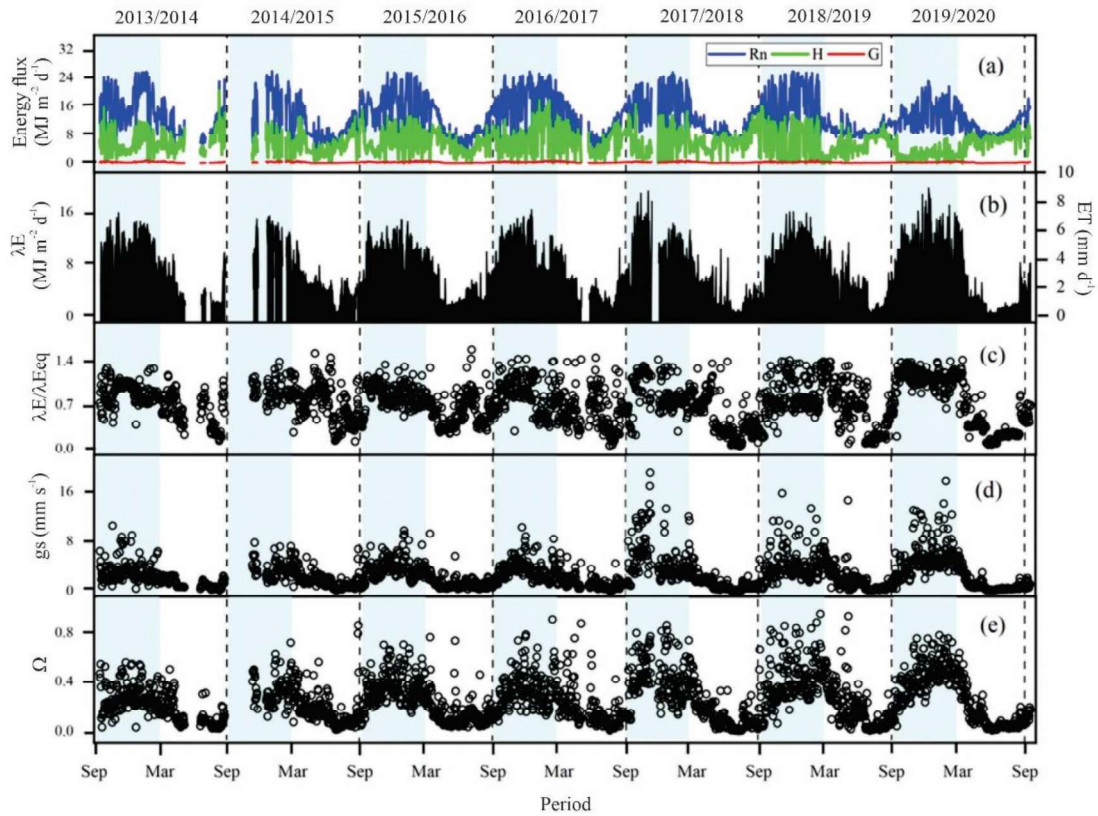


Figure 6. Annual modeled data of (a) daily energy flux: net radiation, sensible heat flux and soil heat flux (Rn, H and G; left scale); (b) latent heat flux (λE , left scale) and equivalent evapotranspiration in millimeters (ET, right scale); (c) daily coefficient of Priestley-Taylor ($\alpha = \lambda E / \lambda E_{eq}$, left scale); (d) daily canopy conductance (g_s , left scale); and (e) daily decoupling factor (Ω , left scale). Shaded columns represent periods of wet seasons.

Figure 7 highlights the relationship between ET_{PM} (Penman-Monteith evapotranspiration) and ET_{WB} (water balance evapotranspiration). The regression line is not significantly different from line 1:1 ($p < 0.05$), indicating a good performance of the modeled evapotranspiration. Moreover, the value of RMSE was 0.41 mm d^{-1} , and the NS coefficient was 0.86, suggesting that the modeled evapotranspiration was close to those from water balance in the dry season.

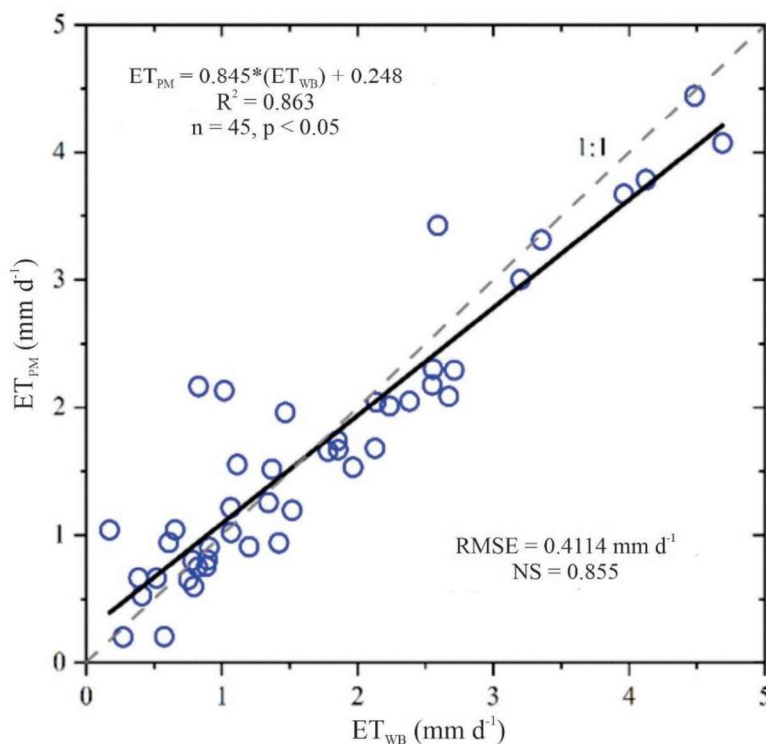


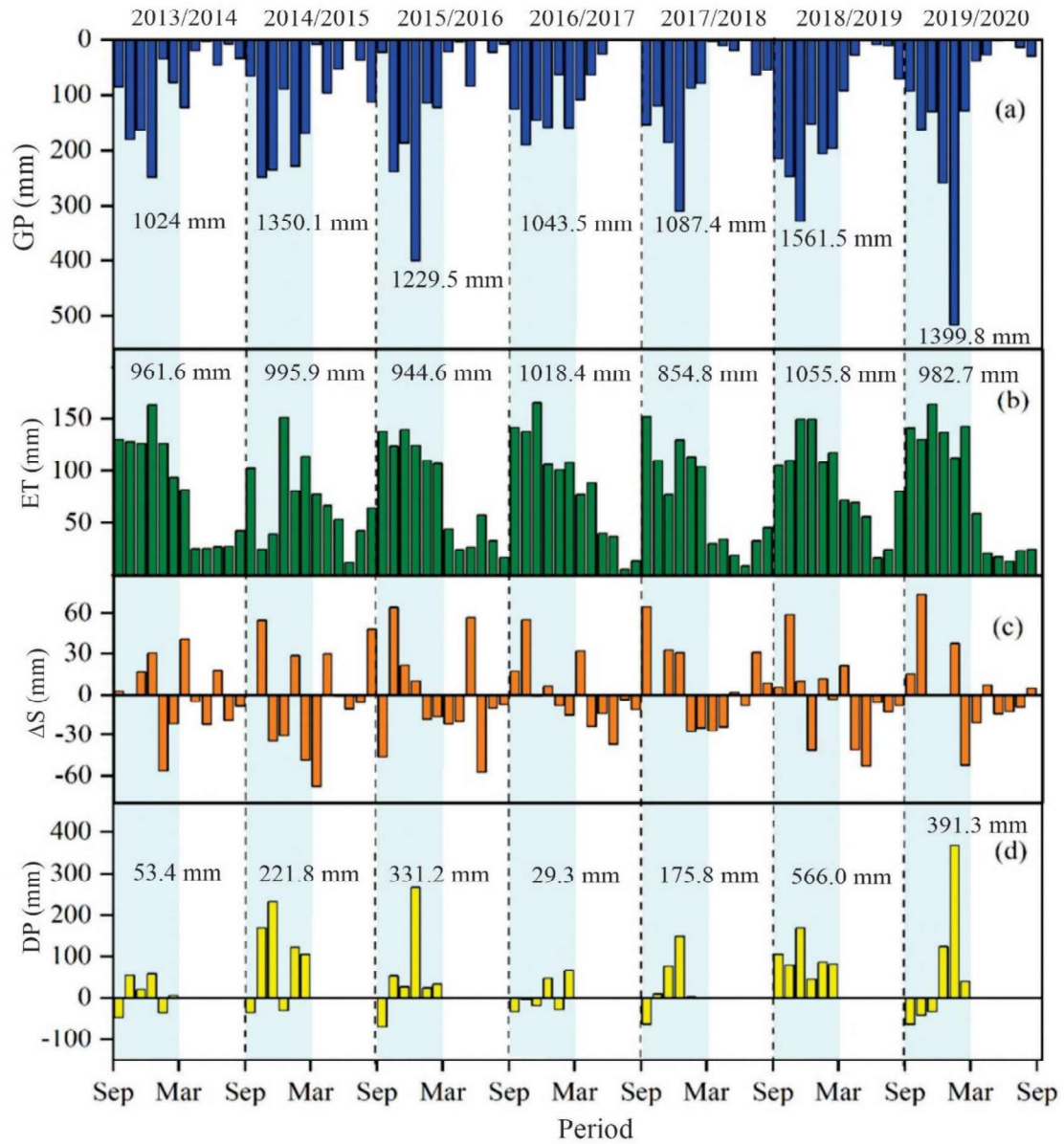
Figure 7. Relationship between modeled evapotranspiration (ET_{PM}) and evapotranspiration from water balance (ET_{WB}) and precision statistics (e.g., RMSE and NS).

Water balance of the SSMF

Figure 8 presents the components of the monthly water balance in SSMF. During the studied period, the maximum values of GP and ET were observed in 2018-2019, with values of 1561.5 and 1055.8 mm, respectively. Notably, monthly ET followed the seasonal pattern of GP, as the summer is the wet season, i.e., the atmospheric demand and REW are higher. The monthly ΔS was predominantly positive in wet periods and negative in dry periods (Figure 8c). Annually, it stands out with positive values in 2016-2017, 2017-2018, and 2019-2020. In the transition between wet and dry periods, ΔS was marked by negative values as evapotranspiration was kept high in response to soil water and energy availability.

Percolation (DP) is predominantly positive on an annual scale, with an average value of 252.7 mm (± 191.5 mm) in the study period. In drier years (2013-2014, 2016-2017), annual DP decreased

significantly (Figure 8d), being only 4% of precipitation, whereas for abnormal/wet years, it was 25% of precipitation. ET showed marked participation in the drier years, which affected the percolation due to higher atmospheric demand and lower than average GP.



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Figure 8. Monthly water balance components: (a) Gross precipitation (GP); (b) evapotranspiration (ET); (c) water storage variation (ΔS) in the root zone; and (d) deep percolation (DP) for the seven hydrological studied years (2014 to 2020).

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Discussion

Meteorological behavior in SSMF

Overall, in these seven hydrological years, the temperature, mainly in summer, was higher than the average, indicating a possible increase in evapotranspiration. The average GP (1242 mm) for these seven years was below the average for the SSMF region compared to the last normal climatology, which is 1383.4 mm (approximately 10% below the average). GP was notably below the average for 2013-2014, 2016-2017, and 2017-2018, resulting in significantly drier conditions. Although the GP values for the last two hydrological years are greater than the average, they were insufficient to fully replenish the soil water content and restore the hydrological processes in the SSMF. Seasonal precipitation patterns in SSMF roughly follow the regional precipitation trends. However, over the past decade, the southeastern region of Brazil has experienced a significant decrease in precipitation compared to the long-term average. The hydrological years of 2013-2014, 2016-2017, and 2017-2018 were characterized by severe to moderate meteorological droughts, particularly during the wet season (Guauque-Mellado et al., 2021). Notably, the 2013-2014 and 2016-2017 hydrological years saw the lowest annual precipitation values in decades, approximately 25% of the long-term average, provoking a change in the temporal distribution of the rainfall in the wet season and a severe reduction in soil water storage. Therefore, the combination of higher temperatures and lower precipitation has provoked a drier condition in the SSMF in the studied period. Our observations over this time (not present here) have shown that the mortality rate of the trees has increased worryingly, making clear the dominance of *Xylopia brasiliensis* over other species. This aspect needs to be deeply studied based on a new forest inventory to confirm this hypothesis, which concerns forests like SSMF losing their biodiversity in the future.

The RWE is an important hydrological indicator because it demonstrates how much water is available in the soil for percolation and evapotranspiration of the forest. Our results indicate that RWE in deeper layers was kept close to 1.0, i.e., trees' root systems slightly uptake water from this depth, focusing more on the shallow layers even in the dry season. Thus, the water balance in the dry season

can be accurately assessed by considering a soil layer confined to 1.0 m without losing accuracy in ET estimates.

It is interesting that ΔS response to GP was more sensitive during the dry season, highlighting positive variations close to (e.g., 2015 and 2018) or even greater than (e.g., 2016) those observed in the wet season. This is due to the drier soil and lower evapotranspiration, which translates into greater storage (empty porous) and the latent response of the forest to the new water availability. The seasonality of ΔS followed variations in GP, corresponding with minor changes in rainfall patterns. Consequently, drought conditions can lead to significant soil water deficits, even during the wet season, due to short dry spells common in the region (Junqueira Junior et al., 2019). For instance, January 2016 and February 2020 exhibited ΔS values comparable to those in drier months. Junqueira et al. (2017) studied the temporal stability of soil moisture in the SSMF from 2012 to 2015. They observed that forest influences soil moisture stability at shallower depths (up to 0.40 m). In contrast, soil attributes, such as saturated hydraulic conductivity, govern moisture behavior in deeper depths. Similarly, Mello et al. (2019) observed comparable results in another Atlantic Forest remnant with shallow soil (Inceptisols). These findings underscore the crucial role of Atlantic Forests in shaping the temporal and spatial behavior of soil moisture, thereby affecting water balance outcomes.

The log relationship between C/GP and GP is relevant because more drainage pathways are formed as the canopy saturates (Rodrigues et al., 2022). As GP increases, canopy interception becomes less critical, and more water is transformed into TF and Sf. This impacts the rainfall redistribution within the forest and the spatial dynamics of soil moisture (Fischer-Bedtke et al., 2023). The variabilities observed in C/GP are due to the biodiversity and structure of the ecosystem, leading to complex canopy properties such as differences in the foliage size, roughness of trunks, and funneling effect, among other vegetative aspects (Terra et al., 2018). In the SSMF, C/GP had an average value in the studied period of 21.06%, a value close to that obtained in an Atlantic Rainforest at the Mantiqueira Mountain Range in southeastern Brazil (20.9%, Mello et al., 2019). Salemi et al. (2013) found up to 33% of this ratio in a neotropical forest at Serra do Mar Mountain Range, also in southeastern Brazil. This difference can

be attributed to the canopy aspects, as in Serra do Mar, an ombrophilous forest was monitored in lower elevations, meaning a canopy with large leaves and a closer canopy over the year.

It is important to highlight the impacts of the severe meteorological droughts on the SSMF hydrological behavior. ET corresponded to 94% and 97.6% of GP in the 2013/2014 and 2016/2017 hydrological years, respectively. For comparison, this ratio varied from 67.6% to 78.6% in the abnormal and wet years. Our study encompasses the two most severe hydrological years observed in the SSMF, which were caused by negative anomalies in precipitation and positive in temperatures, increasing ET. In addition, the deep percolation in these years was, respectively, 53.4 mm and 29.3 mm. Thus, SSMF faced a climate condition that has been a challenge for the trees' survival, compromising the forest biodiversity, and can hit the region throughout the XXI Century (Silva and Mello, 2021).

Evapotranspiration modeling

The evapotranspiration in SSMF showed a relevant variability, which can be attributed to the spatial distribution of water inputs in the soil, such as canopy heterogeneity, and the distribution of soil physical properties, like soil saturated hydraulic conductivity. Overall, evapotranspiration variability in SSMF is low, primarily due to the uniform physical properties of the Red Latosol (spatially and in-depth), which also results in low variability of the REW during the dry season. Evapotranspiration estimates in this study are higher than the findings from prior research in the Atlantic Forest ecosystem. For example, Pereira et al. (2010), Salemi et al. (2013), and Mello et al. (2019) reported daily average values of 3.9, 3.56, and 3.26 mm d⁻¹, respectively, using only two and wetter hydrological years, thus, little conclusive results. In our study, daily evapotranspiration varies from 4.96 mm d⁻¹ in the wet season to 1.55 mm d⁻¹ in the dry season, showing the relevance of taking into account longer periods of analysis so that the forest faced with drier and wetter hydrological years.

The exponential relationship between ET and GP (Figure 4d) highlights the transition from a water-limited environment to an energy-limited environment. This process could be detected when analyzing the differences in the evapotranspiration of the wet (blue dots) and dry periods (green dots) (Figure 4d). Evapotranspiration increases with GP due to higher REW, which enhances stomata opening

and gas exchange during photosynthesis. A stabilization is reached for significant amounts of precipitation from which REW is no longer the limiting factor for evapotranspiration.

The meteorological data and estimated canopy conductance support the evapotranspiration modeling approach for SSMF (i.e., a semi-deciduous forest). The Penman-Monteith method could adequately estimate the actual evapotranspiration of the SSMF, showing the clear seasonality of the meteorological elements that mainly explain the ET behavior. The implementation of the evapotranspiration model accurately simulated the observed values from the water balance analysis in the dry seasons, exhibiting a deviation of 13.7%. This error margin aligns with findings from Mello et al. (2019), Muñoz-Villers et al. (2012), and Pereira et al. (2010).

The model facilitated an investigation into the seasonal variations in evapotranspiration observed in the forest over seven hydrological years, predominantly driven by fluctuations in radiation and soil moisture levels. Specifically, radiation accounted for 60% of the evapotranspiration responses, a finding consistent with other studies conducted in Neotropical forests across Brazil and Mexico, particularly in semi-deciduous forest sites (Cabral et al., 2015; Rocha et al., 2004; Muñoz-Villers et al., 2012).

The gas exchange rates observed during the study aligned with those reported for secondary tropical forests, exhibiting average daily values ranging from 0.9 mm s⁻¹ to 12.7 mm s⁻¹ (Cabral et al., 2015; Rocha et al., 2004; Fraga et al., 2015; Marques et al., 2020). Tan et al. (2019) noted that in tropical forests, variations in precipitation substantially impact gas exchange, i.e., photosynthesis and transpiration, particularly during the dry season, as a means for the forest to mitigate water and energy loss. This phenomenon was evident in our study, which showed a marked decline in canopy conductance from March to September, i.e., in the dry season. Furthermore, we observed that evapotranspiration rates increase exponentially in response to changes in vapor pressure deficit (VPD), particularly when VPD exceeds 0.7 kPa (Figure 1S – supplementary). However, the gas exchange decreases with VPD, showing a stomatal regulation due to the environmental conditions (i.e., increased atmospheric demand). This forest mechanism regulates the water transfer to the atmosphere, controlling the root water uptake and soil water depletion. Corroborating our findings, Cabral et al. (2015) and

Igarashi et al. (2015) documented peak evapotranspiration rates preceding minimal VPD levels during rainy periods, a result of enhanced stomatal opening driven by the water vapor transport gradient.

In terms of biophysical evapotranspiration controls, the seasonal variations of the decoupling factor (Ω) and Priestly-Taylor coefficient ($\alpha = \lambda E / \lambda E_{eq}$) were more pronounced in the rainy period (Ma et al., 2015; Tan et al., 2019). The results suggest that in SSMF, stomatal regulation predominantly governs evapotranspiration. The distribution of α (Figure 6c) showed a greater atmospheric demand in wet periods than in dry periods, influencing stomatal response to water availability. This relationship between α and relative extractable water (REW) across five soil depths was clearly demonstrated in Figure 2S – supplementary). Notably, water depletion occurred more rapidly in shallower layers (0-0.2 m). Maseyk et al. (2008), Aguilos et al. (2019), and Jiao et al. (2019) mentioned that evapotranspiration is strongly constrained by soil moisture, particularly under water deficits that lead to decreased canopy conductance to mitigate water stress from soil moisture depletion. This mechanism is crucial in ecosystems like the Atlantic Forest, which experiences seasonal climate variation and periods of reduced precipitation. Therefore, more extended drought periods, potentially due to climate change, can significantly impact water availability in the soil, thereby, affecting evapotranspiration and photosynthesis.

Highlights of the water balance

The analysis of water balance elements revealed significant variances across the seven hydrological years studied. Annually, the capillary effect was shown to be negligible, even under severe drought conditions (Figure 8d). However, in months where ET exceeds GP, DP becomes negative. This occurs mainly in months after the dry period (September and October), where evapotranspiration tends to increase due to increase in radiation. Therefore, small changes in rainfall can reduce the water percolation in this ecosystem. In addition, our results, aligned with those of other studies (e.g., Li et al., 2010; Negrón Juárez et al., 2007), suggest that deep soil water storage in the previous wet season generally sustains forest evapotranspiration during subsequent dry periods, thereby mitigating the

effects of rainfall deficits. The temporal behavior of REW at 1.0 m in depth has little variability, suggesting that the water stored in this layer suffices to support the ecosystem in the dry period. The highest DP values followed those of GP, primarily in the middle of the wet season. In these moments, REW was high (due to previous rains), and evapotranspiration was driven mainly by energy availability. Therefore, excess water ($\theta > \theta_{FC}$) was transported throughout the profile by activation/optimization of preferential pathways, increasing DP. This suggests efficient macropore drainage as saturation was not reached with increased GP (Jiang et al., 2023), driving water to deeper layers.

The Atlantic Forest is critical in regulating water use during drought conditions. No significant water deficit was observed in the SSMF over the study period, even during the dry years. However, drought conditions did impact the SSMF's hydrological processes by reducing canopy interception, soil moisture, and percolation rates (Guauque-Mellado et al., 2021). On the other hand, the variability in evapotranspiration did not align with the growth patterns typically seen in secondary tropical forests, a phenomenon reported in previous researches (e.g., Kume et al., 2011; Von Randow et al., 2020). This disparity may be attributed to (i) the edge effect, which hinders forest growth and diversification; (ii) climatic variability; and (iii) the trees' physiological strategies to cope with water deficit.

This study presents some limitations and uncertainties (i) ΔS in finer temporal resolution as a monthly step precludes the analysis of the temporal variability in shorter scales; (ii) the soil layer up to 1.0 m depth, which is a source of uncertainty for percolation estimates as some tropical species can reach deeper layers (Rocha et al., 2004; Kume et al., 2011; Broedel et al., 2017); (iii) measurement of ET using a more precise approach, such as eddy covariance and sap flow. Despite these shortcomings, the analysis highlights the importance of evapotranspiration in the SSMF, conducted in a forest site in Brazil for the first time throughout seven consecutive hydrological years, under different climate patterns. These results could improve the understanding of tropical forests under climate change and support decision-makers on better management practices for Atlantic Forest biomes.

Conclusions

The study delineates evapotranspiration as a pivotal component of the water balance in Atlantic Forest biomes, exhibiting spatial variability and critical linkage with factors such as soil moisture, vapor pressure deficit, air temperature, and atmospheric pressure. This process is essential in regulating hydrological mechanisms, particularly soil water storage during periods marked by water scarcity. The primary insights from the investigation are outlined as follows:

- a) On average, net precipitation (TF + Sf) accounted for 78.9% of GP, comprising 99.8% throughfall and 0.2% streamflow. Canopy interception was significant in the forest and corresponded to 21% of GP over the studied period.
- b) In the driest hydrological years (2013/2014; 2016/2017), evapotranspiration constituted more than 94% of GP, threatening species not adapted to this change in the climate pattern.
- c) Throughout the studied period, most hydrological years ended with negative soil water storage changes at 1.0 m in depth, except for the 2018/2019 and 2019/2020 hydrological years.
- d) The study observed minimal interannual spatial variations in evapotranspiration at the SSMP (CV=4.72%). Evapotranspiration estimates were reliable across different scales, enhancing the understanding of water balance within the forest ecosystem.
- e) Evapotranspiration accounted for 60% of Rn. During the dry seasons, it was primarily governed by soil water availability and stomatal regulation, whereas in wet seasons, the available energy and atmospheric conditions largely influenced its behavior.
- f) Water percolation below the root zone showed no significant capillary rise, mainly because of the depth of the Oxisols. It varied from 4% and 25.4% of GP for severe drought years and abnormal/wet hydrological years, respectively.

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Supplementary

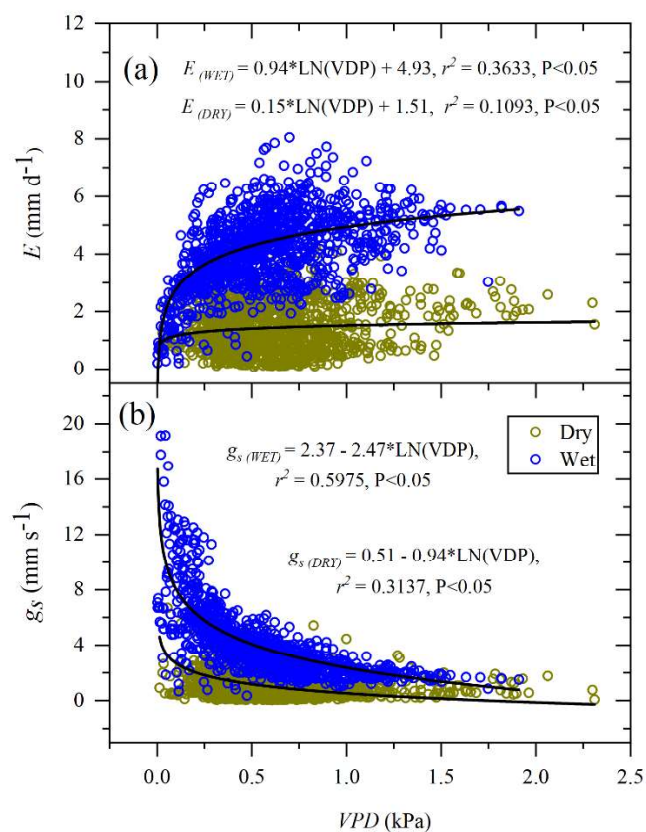


Figure 1S. Relationships between vapor pressure deficit (VPD) with (a) daily modeled evapotranspiration and (b) daily g_s .

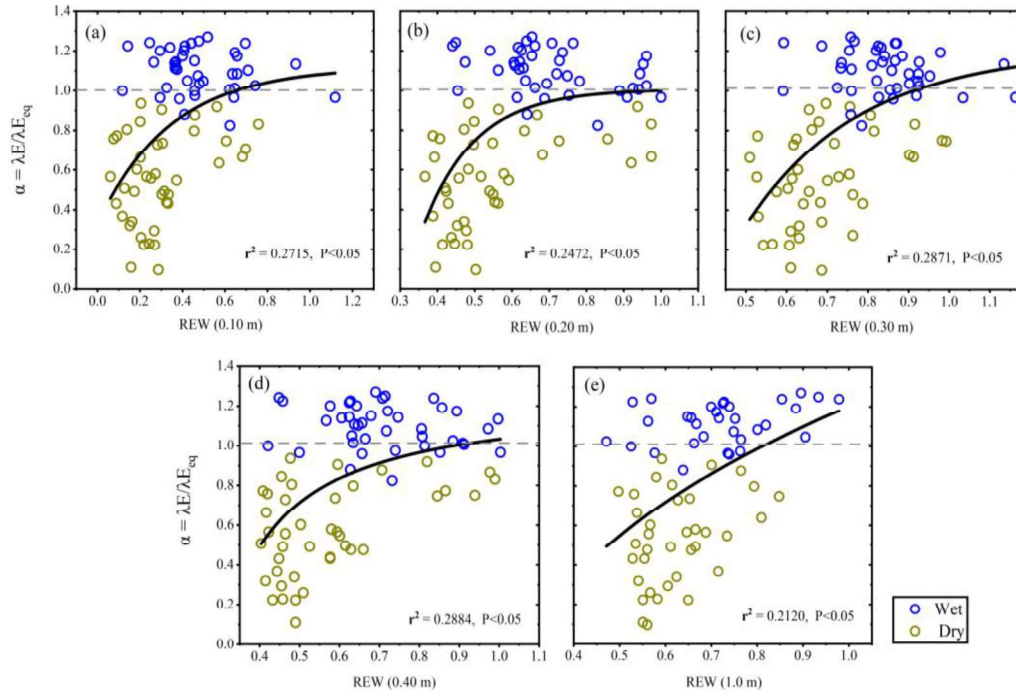


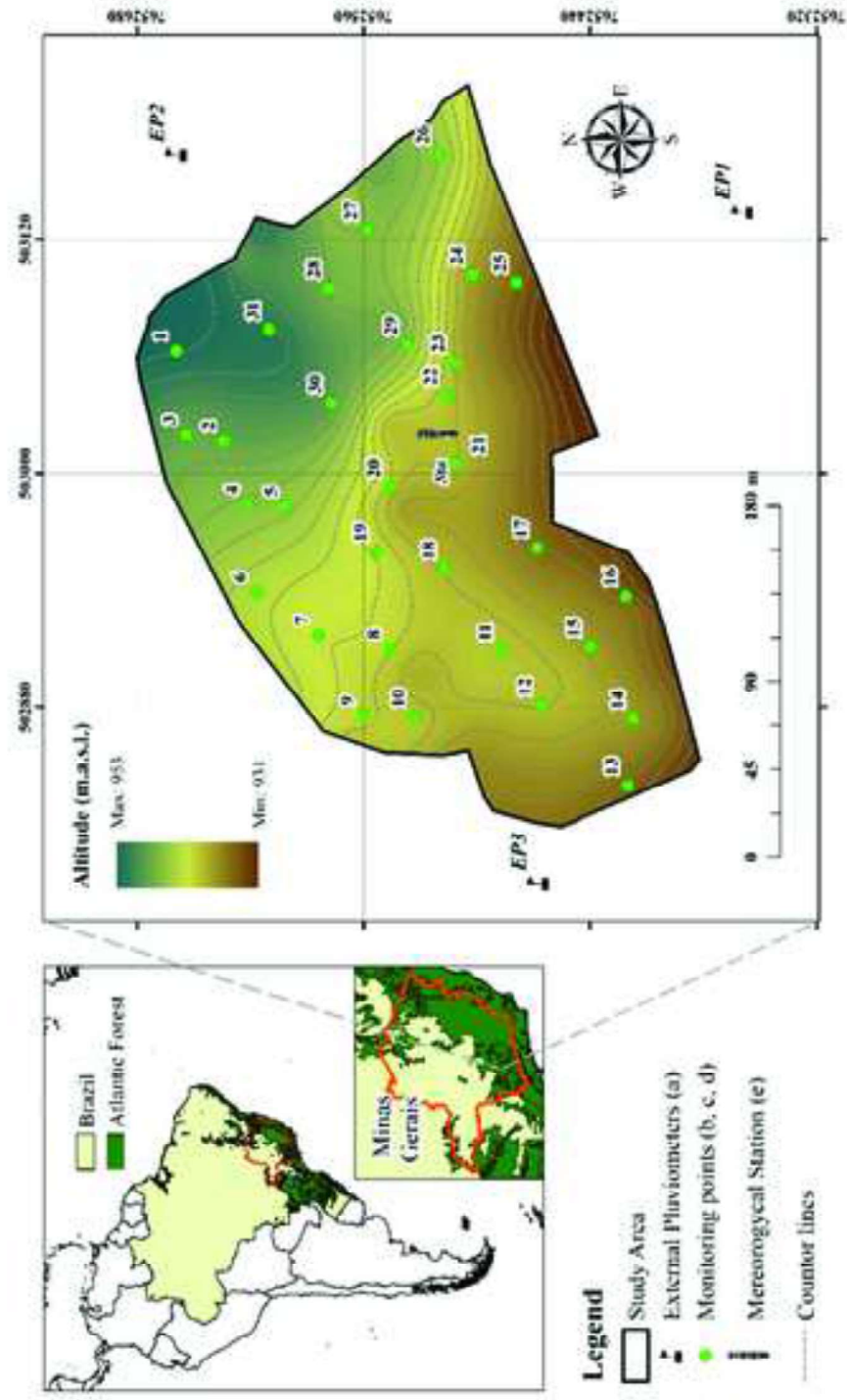
Figure 2S. Relationships between the monthly mean values of the Priestley-Taylor coefficient ($\alpha = \lambda E / \lambda E_{eq}$) and the relative extractable soil water content (REW) (a. 0.10 m; b. 0.20 m; c. 0.30 m; d. 0.40 m; and e. 1.0 m).

Table A1. Daily average of gross precipitation (GP, mm d⁻¹), throughfall (TF, mm d⁻¹), streamflow (St), and canopy rainfall interception (C) in the dry and wet periods of the hydrological years from 2014 to 2020.

Hydrological Year		GP (mm.d ⁻¹)	TF (mm.d ⁻¹)	St (mm.d ⁻¹)	C (mm.d ⁻¹)
2013-2014	Wet	4.13 ± 2.7	3.19 ± 2.1	0.01 ± 0.02	0.85 ± 0.6
	Dry	0.9 ± 1.5	0.62 ± 1.3	0.01 ± 0.01	0.18 ± 0.2
	Annual	2.51 ± 2.1	1.9 ± 1.7	0.01 ± 0.01	0.52 ± 0.4
2014-2015	Wet	6.67 ± 2.6	5.3 ± 2.2	0.02 ± 0.02	0.87 ± 0.8
	Dry	1.49 ± 1.5	1.34 ± 1.4	0.01 ± 0.01	0.24 ± 0.2
	Annual	4.08 ± 2.1	3.32 ± 1.8	0.01 ± 0.01	0.56 ± 0.5
2015-2016	Wet	5.16 ± 4.3	4.41 ± 3.7	0.01 ± 0.02	1.08 ± 1.2
	Dry	0.51 ± 1.0	0.4 ± 0.51	0.01 ± 0.01	0.13 ± 0.6
	Annual	2.84 ± 2.7	2.4 ± 2.1	0.01 ± 0.01	0.61 ± 0.9
2016-2017	Wet	5.05 ± 1.4	4.07 ± 1.3	0.01 ± 0.01	0.98 ± 0.4
	Dry	0.46 ± 1.5	0.36 ± 1.3	0.01 ± 0.00	0.12 ± 0.1
	Annual	2.76 ± 1.5	2.21 ± 1.3	0.01 ± 0.00	0.54 ± 0.3
2017-2018	Wet	4.54 ± 2.8	4.28 ± 1.8	0.01 ± 0.01	0.45 ± 1.2
	Dry	0.51 ± 0.9	0.33 ± 0.7	0.01 ± 0.00	0.18 ± 0.2
	Annual	2.53 ± 1.9	2.3 ± 1.3	0.01 ± 0.00	0.31 ± 0.7
2018-2019	Wet	7.01 ± 2.0	5.73 ± 1.7	0.01 ± 0.01	1.47 ± 0.4
	Dry	0.66 ± 1.2	0.43 ± 1.0	0.01 ± 0.00	0.23 ± 0.2
	Annual	3.83 ± 1.6	3.08 ± 1.4	0.01 ± 0.00	0.85 ± 0.3
2019-2020	Wet	4.86 ± 5.3	3.73 ± 4.9	0.01 ± 0.02	1.20 ± 1.2
	Dry	0.68 ± 0.5	0.44 ± 0.4	0.01 ± 0.01	0.21 ± 0.1
	Annual	2.77 ± 2.9	2.08 ± 2.6	0.01 ± 0.01	0.72 ± 0.7

Table A2. Daily average values of net radiation (R_n , MJ m⁻² d⁻¹), latent heat flux (LE, MJ m⁻² d⁻¹), sensible heat flux (H, MJ m⁻² d⁻¹), soil heat flux (G, MJ m⁻² d⁻¹), latent heat flux (λE_{res} , MJ m⁻² d⁻¹), coefficient of Priestley-Taylor ($\alpha = \lambda E / \lambda E_{eq}$), canopy conductance (g_s , mm s⁻¹), decoupling factor (Ω) and evapotranspiration (E, mm d⁻¹) in the dry, wet and annual time scales of the hydrological years from 2014 to 2020.

Hydrological Year		R_n	H	G	λE_{res}	A	g_s	Ω	E
2013-2014	Wet	15.89 ± 5.1	5.38 ± 3.1	0.45 ± 0.2	10.64 ± 2.9	0.89 ± 0.2	2.29 ± 1.6	0.25 ± 0.1	4.33 ± 1.2
	Dry	9.44 ± 3.3	5.62 ± 3.1	0.2 ± 0.1	3.26 ± 2.2	0.57 ± 0.3	0.69 ± 0.7	0.09 ± 0.1	1.33 ± 0.9
	Annual	12.67 ± 4.2	5.5 ± 3.1	0.32 ± 0.1	6.95 ± 2.5	0.73 ± 0.2	1.49 ± 1.2	0.17 ± 0.1	2.83 ± 1.0
2014-2015	Wet	17.48 ± 5.9	6.09 ± 3	0.47 ± 0.2	10.49 ± 3.4	0.9 ± 0.2	3.39 ± 1.5	0.31 ± 0.1	4.29 ± 1.4
	Dry	8.61 ± 3.8	4.95 ± 3.2	0.17 ± 0.1	4.3 ± 1.60	0.63 ± 0.3	1.34 ± 1	0.14 ± 0.1	1.75 ± 0.7
	Annual	13.04 ± 4.9	5.52 ± 3.1	0.32 ± 0.1	7.39 ± 2.5	0.77 ± 0.2	2.36 ± 1.2	0.22 ± 0.1	3.02 ± 1.0
2015-2016	Wet	16.95 ± 4.7	5.6 ± 3.1	0.42 ± 0.2	10.05 ± 2.3	0.86 ± 0.2	3.34 ± 1.5	0.33 ± 0.1	4.1 ± 0.90
	Dry	8.87 ± 3.8	6.26 ± 2.6	0.17 ± 0.1	2.82 ± 2.1	0.55 ± 0.2	0.88 ± 0.6	0.11 ± 0.1	1.15 ± 0.8
	Annual	12.91 ± 4.3	5.93 ± 2.8	0.29 ± 0.1	6.43 ± 2.2	0.7 ± 0.2	2.11 ± 1.1	0.22 ± 0.1	2.63 ± 0.9
2016-2017	Wet	18.71 ± 5.1	7.06 ± 4.2	0.45 ± 0.2	9.63 ± 3.0	0.87 ± 0.3	3.17 ± 1.6	0.31 ± 0.1	3.93 ± 1.3
	Dry	10.1 ± 3.9	7.24 ± 3.2	0.19 ± 0.1	3.44 ± 2.1	0.57 ± 0.3	1.04 ± 1.0	0.12 ± 0.1	1.4 ± 0.8
	Annual	14.4 ± 4.5	7.15 ± 3.7	0.32 ± 0.1	6.53 ± 2.6	0.72 ± 0.3	2.11 ± 1.3	0.22 ± 0.1	2.67 ± 1.0
2017-2018	Wet	15.56 ± 4.7	5.53 ± 3.6	0.45 ± 0.2	10.42 ± 3.3	0.84 ± 0.2	3.75 ± 3.7	0.41 ± 0.2	4.26 ± 1.4
	Dry	9.08 ± 3.2	6.51 ± 3.1	0.17 ± 0.1	2.29 ± 2.0	0.33 ± 0.3	0.6 ± 0.7	0.07 ± 0.1	0.94 ± 0.8
	Annual	12.32 ± 3.9	6.02 ± 3.4	0.31 ± 0.1	6.36 ± 2.7	0.59 ± 0.3	2.18 ± 2.2	0.24 ± 0.1	2.6 ± 1.10
2018-2019	Wet	15.82 ± 5.9	5.76 ± 4.3	0.4 ± 0.2	10.1 ± 3.0	0.81 ± 0.3	3.62 ± 2.4	0.38 ± 0.2	4.13 ± 1.3
	Dry	9.07 ± 1.5	5.33 ± 2.3	0.18 ± 0.1	3.78 ± 2.3	0.62 ± 0.3	0.91 ± 1.1	0.12 ± 0.1	1.54 ± 0.9
	Annual	12.45 ± 3.7	5.54 ± 3.3	0.29 ± 0.1	6.94 ± 2.6	0.71 ± 0.3	2.27 ± 1.7	0.25 ± 0.2	2.83 ± 1.1
2019-2020	Wet	13.1 ± 3.5	2.2 ± 1.3	0.34 ± 0.1	10.66 ± 3.1	1.15 ± 0.1	4.71 ± 2.7	0.45 ± 0.1	4.35 ± 1.3
	Dry	9.36 ± 2.7	7.22 ± 1.7	0.18 ± 0.1	1.91 ± 1.9	0.27 ± 0.2	0.48 ± 0.6	0.06 ± 0.1	0.78 ± 0.8
	Annual	11.23 ± 3.1	4.71 ± 1.5	0.26 ± 0.1	6.29 ± 2.5	0.71 ± 0.2	2.59 ± 1.6	0.25 ± 0.1	2.57 ± 1.0



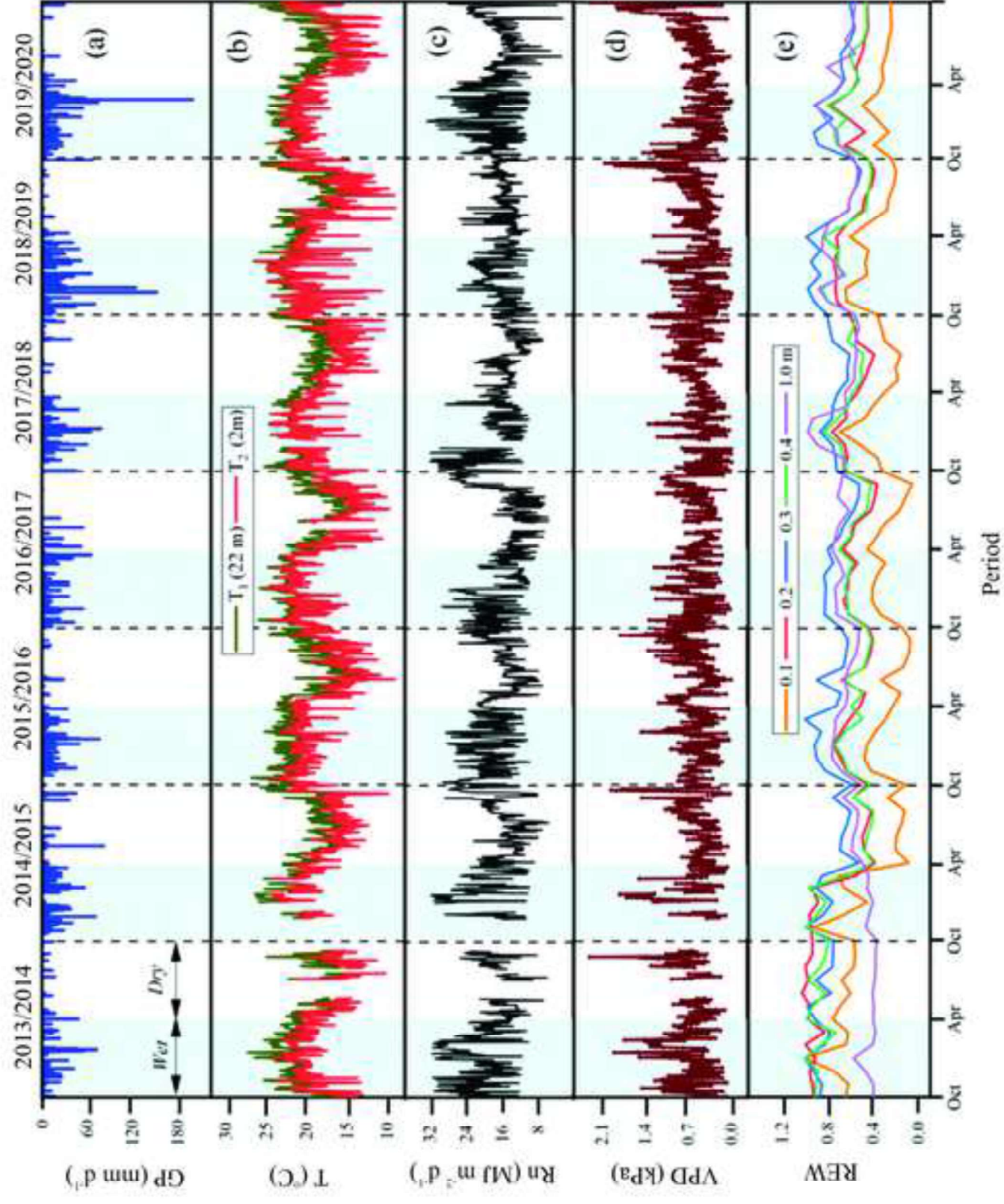


Figure 2

Figure 3

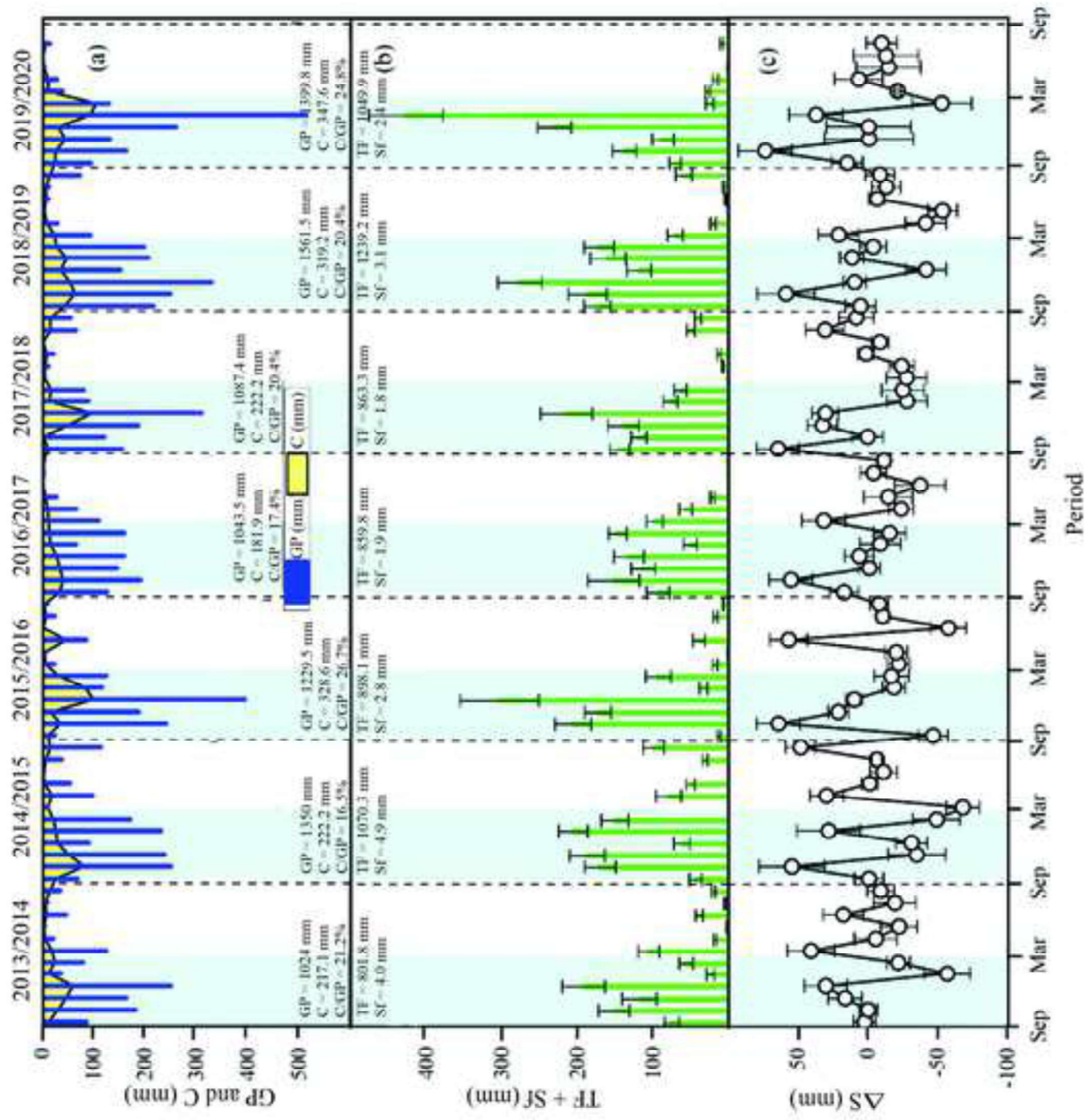
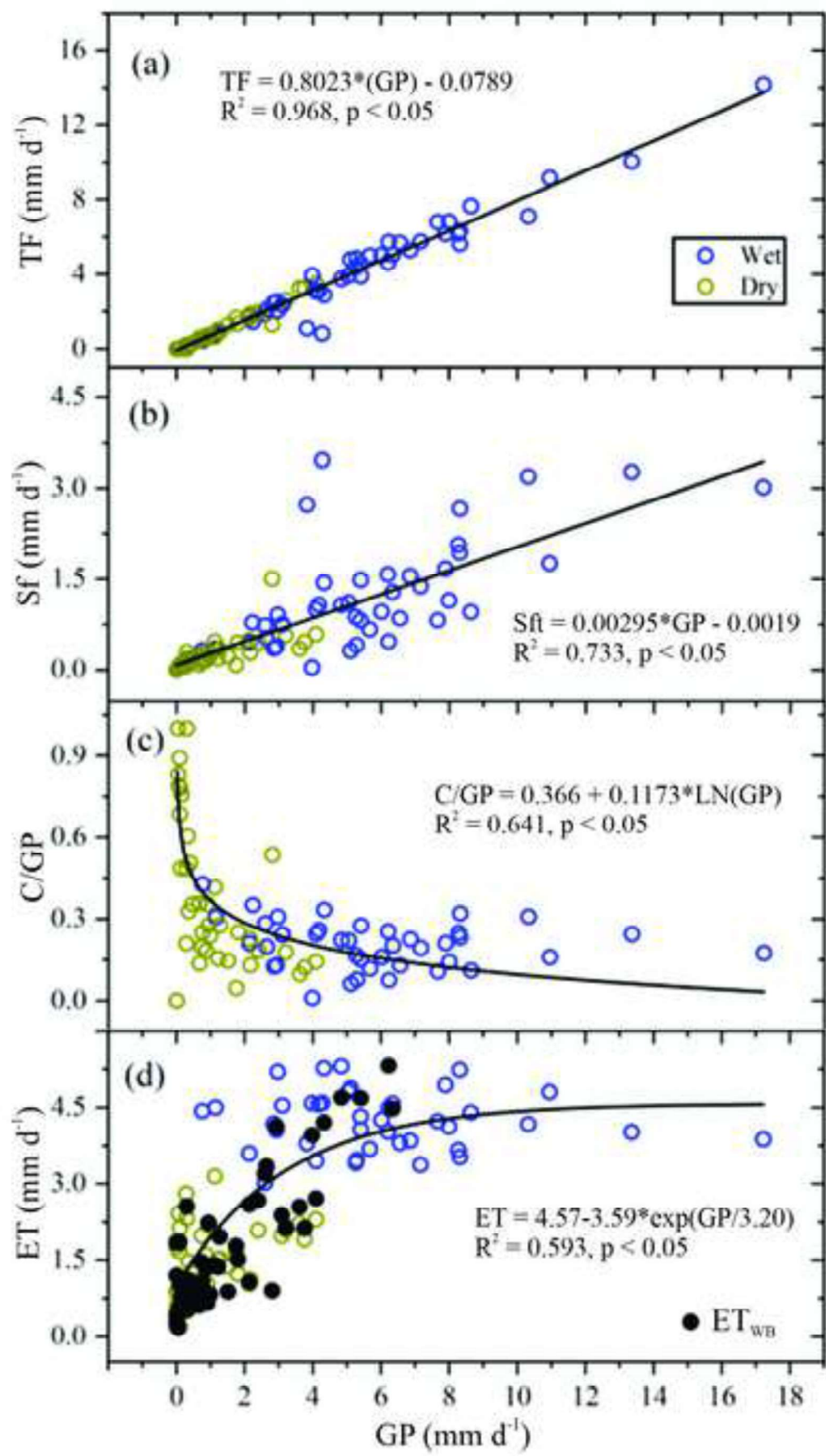


Figure 4

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(i) Multiannual variability of actual evapotranspiration (ET)

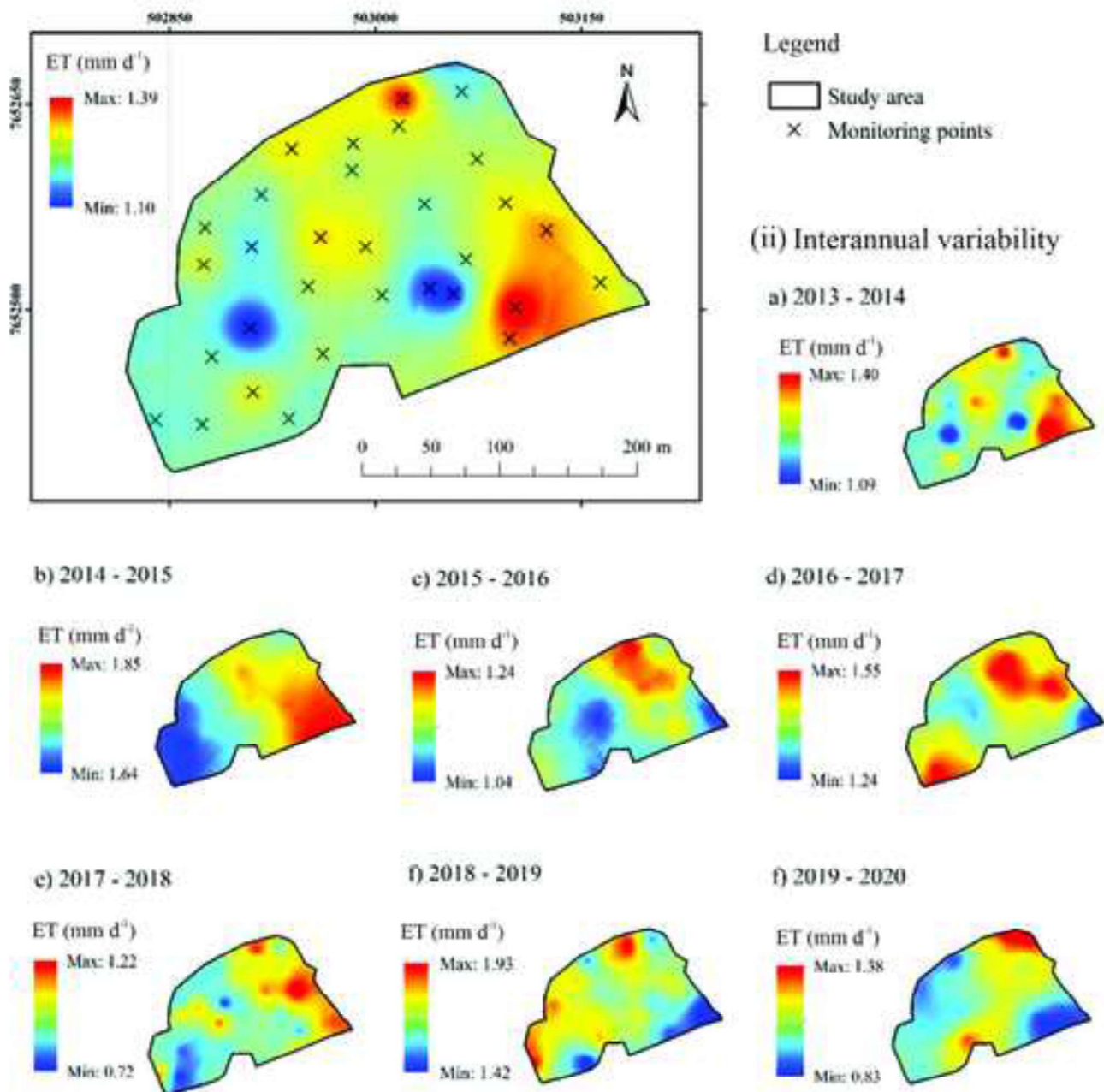


Figure 6

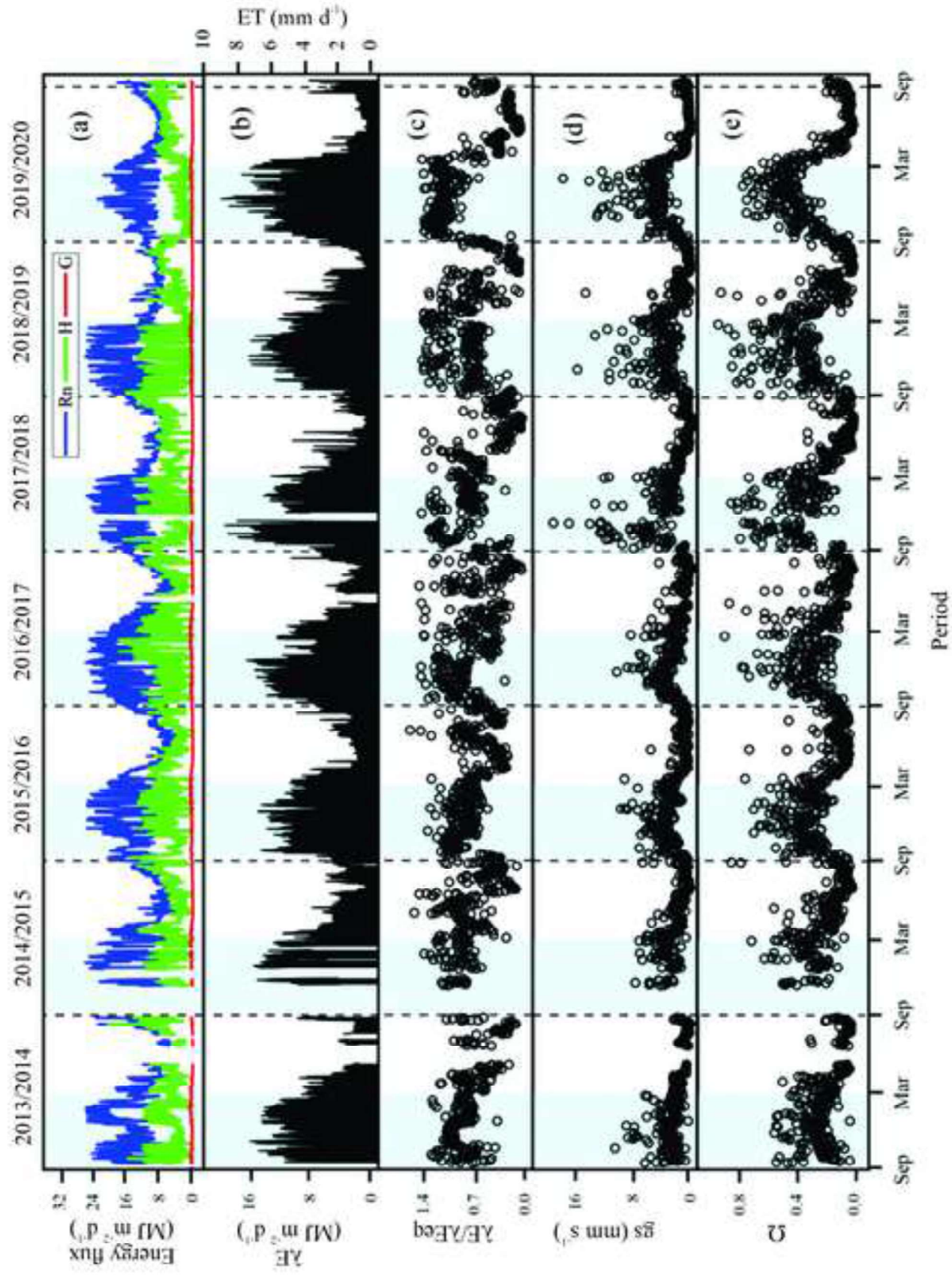
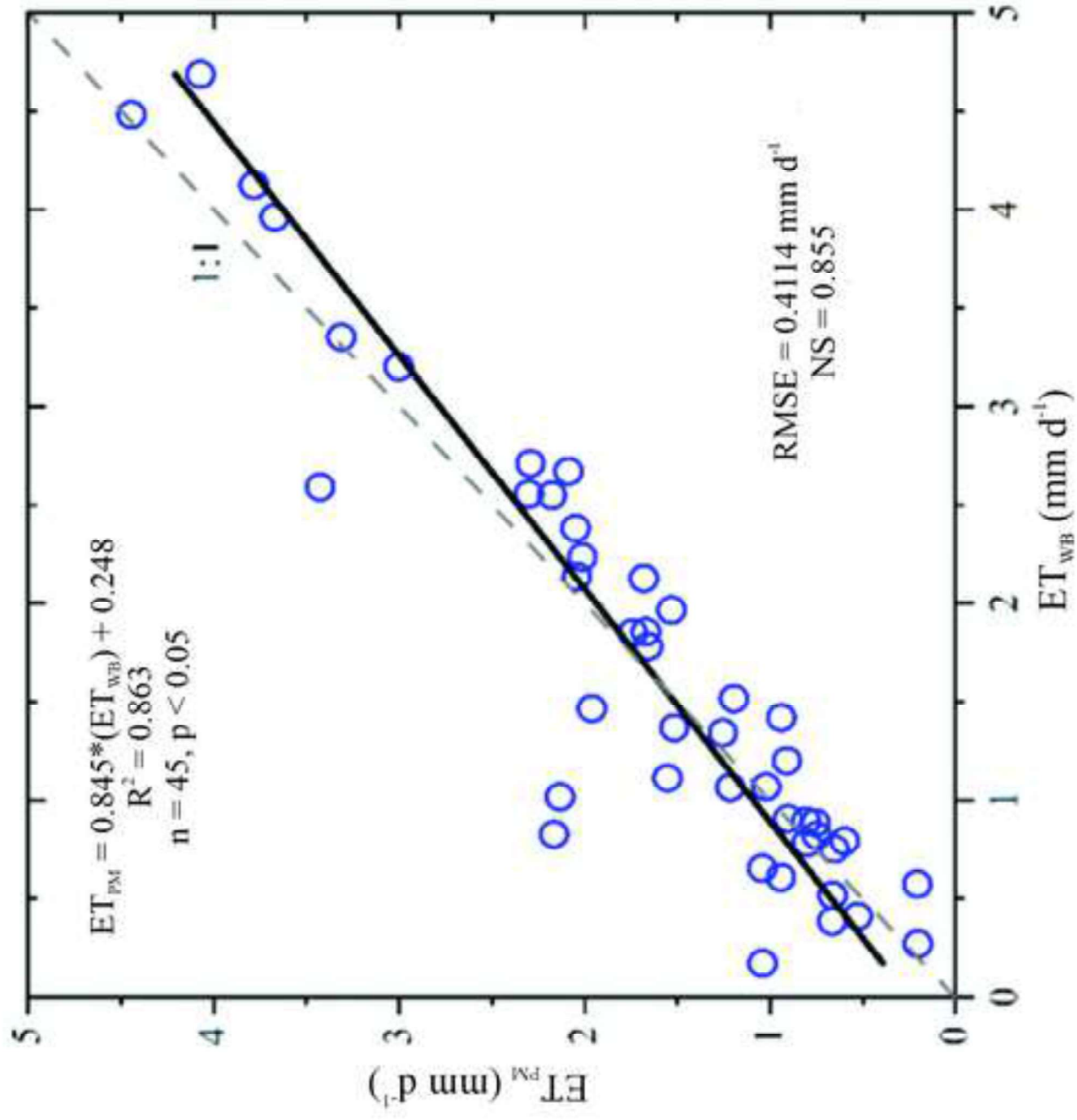
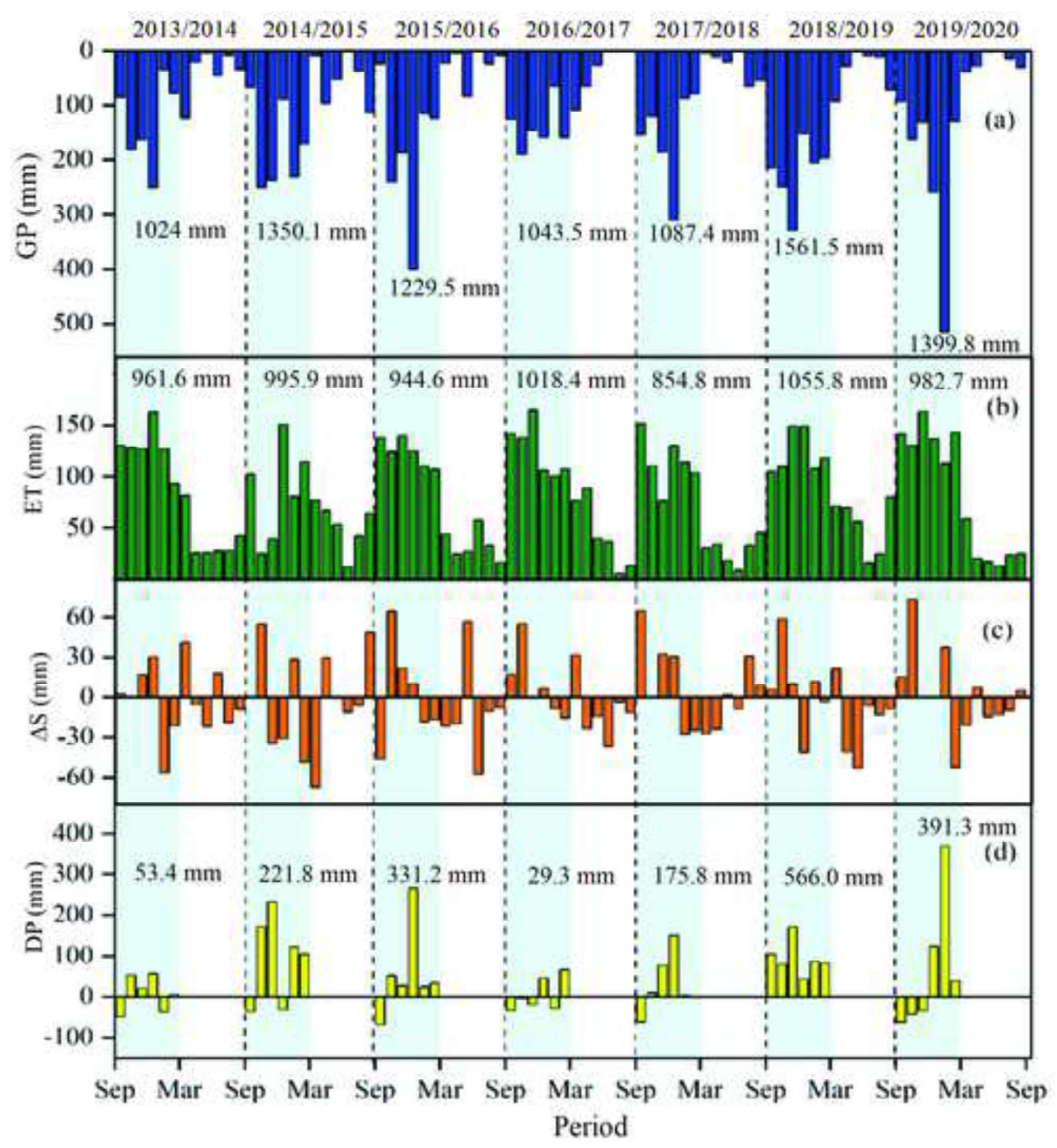


Figure 7





Declaration of interests

☒ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐ The author is an Editorial Board Member/Editor-in-Chief/Associate Editor/Guest Editor for *[Journal name]* and was not involved in the editorial review or the decision to publish this article.

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